

ESSENTIAL MATHEMATICS GRADE 10: TRIGONOMETRY

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.3 (8 Lessons)

ATTRIBUTION AND LICENSING

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SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.3: Trigonometry
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Identification of sides of a right-angled triangle (hypotenuse, opposite, adjacent) relative to a specific acute angle – Sine, cosine and tangent ratios of acute angles in a right-angled triangle – Sines and cosines of complementary angles and their relationship – Application of trigonometric ratios to angles of elevation – Application of trigonometric ratios to angles of depression – Construction and use of a simple clinometer to measure angles of elevation and depression – Use of trigonometric tables and digital tools to find trigonometric ratios – Real-life applications of trigonometry in the immediate environment
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - identify the sides of a right-angled triangle in relation to a particular angle, - determine the sine, cosine and tangent of acute angles in a right-angled triangle, - relate sines and cosines of complementary angles, - apply trigonometric ratios to angles of elevation and depression, - recognise the use of trigonometry in real-life situations.

Core Competencies	– Self-efficacy: The learner makes observations and establishes the relationships between the sine and cosine ratios of complementary angles, building confidence in their own mathematical reasoning as they generate and verify the pattern independently. – Learning to learn: The learner learns collaboratively while constructing a simple clinometer using locally available materials (cardboard, straw, string, protractor) and using it as a hands-on learning resource for measuring angles of elevation and depression in the school environment.
Core Values	– Love: The learner demonstrates a caring attitude towards others when using a simple clinometer to measure the angle of elevation and depression of different objects in the immediate environment, ensuring every group member has a turn and that measurements are taken safely and considerately. – Responsibility: The learner takes care of instruments such as rulers, geometrical sets, graph paper and digital devices used during trigonometry activities, returning them in good condition and ensuring shared resources are available for peers.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners articulate what information is needed (angle, known side) to find an unknown height or distance, framing trigonometry as a tool for answering real measurement questions about their environment. – Planning and Carrying Out Investigations: Learners design and carry out a clinometer investigation — identifying objects to measure, deciding on measurement positions, recording angle and distance data, and checking results for consistency. – Using Mathematics and Computational Thinking: Learners select and apply the appropriate trigonometric ratio (sin, cos, tan) to compute unknown sides and angles, evaluate answers for reasonableness, and use digital tools (calculators, online resources) to verify computations. – Constructing Explanations: Learners use their generated table of complementary angles and their sine/cosine values to construct a reasoned explanation for why $\sin \theta = \cos(90^\circ - \theta)$, connecting numerical patterns to geometric reasoning. – Obtaining, Evaluating and Communicating Information: Learners present clinometer findings and trigonometric solutions to peers, critique each other's methods, and evaluate the accuracy of real-world measurements against calculated values.
Pertinent & Contemporary Issues (PCIs)	– Self-esteem: The learners acquire and strengthen self-esteem while designing and making a simple clinometer using locally available materials (cardboard, string, protractor, drinking straw) and successfully using it as a functional learning resource for measuring real angles in their school environment. – Life Skills (Critical Thinking and Problem Solving): Learners apply trigonometric ratios to authentic measurement problems — such as estimating the height of a flagpole, water tank, or school building in Nairobi — developing analytical and decision-making skills transferable to daily life. – Education for Sustainable Development: Learners use locally sourced, low-cost materials to construct the clinometer, reinforcing resourcefulness and environmental awareness while achieving a meaningful mathematical learning outcome.
Career Connections	– Surveying and Land Management: Trigonometric ratios are foundational tools for surveyors who map land parcels across Kenya's counties — from the Rift Valley to the Coast — measuring distances and elevations using clinometers and theodolites. – Architecture and Building Construction: Architects and construction engineers in Kenya use angles of elevation, roof pitch calculations and trigonometric ratios when designing and constructing buildings, ensuring structural stability. – Aviation and Navigation: Pilots and navigators at airports such as JKIA (Nairobi) and Moi International Airport (Mombasa) apply trigonometry to calculate flight approach angles, altitudes and distances. – Sports Coaching and Biomechanics: Sports scientists and coaches working with Kenyan athletes (e.g., track and field at Kipchoge Keino Stadium, Eldoret) use angle analysis to optimise throwing, jumping and running techniques. – Photography and Media Technology: Photographers and videographers computing camera angles, framing shots and calculating perspective use principles directly related to trigonometric ratios.

	– Forestry and Environmental Science: Forest officers in Kenya's Aberdare and Mau Forest complexes use clinometers to measure tree heights and terrain slopes — a direct application of angles of elevation.
Focus for Lessons	This unit introduces learners to trigonometry through a practical, investigation-led approach anchored in the Kenyan built and natural environment. Learners move from identifying the sides of a right-angled triangle, through deriving and applying the three trigonometric ratios, to discovering the complementary-angle relationship and finally deploying their knowledge to solve real problems involving angles of elevation and depression using a student-built clinometer. Throughout the 8 lessons, every concept is connected back to the anchoring phenomenon of estimating the height of an inaccessible structure using only a measured distance and an angle, reinforcing trigonometry as a practical, empowering tool rather than an abstract set of rules.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment — without ever measuring it directly? KICD KEY INQUIRY QUESTION: 1. Why is trigonometry important?
Anchoring Phenomenon	ANCHORING PHENOMENON — "How tall is the water tank without climbing it?" At many Kenyan schools and homesteads, a large elevated water tank (sometimes mounted on a steel frame 4–8 metres high) is a familiar sight. Students can clearly see the tank from the ground but cannot climb it or use a tape measure along its height directly. The teacher presents this puzzling scenario: Two students stand at different positions on the ground, both a known horizontal distance from the base of the water-tank tower. Each student uses a simple angle-measuring device (a protractor and string) to sight the top of the tank and records a different angle. Without any additional climbing or measuring, can we use ONLY these two pieces of information — the horizontal distance and the measured angle — to calculate exactly how high the top of the tank is above the ground? This is puzzling because: (1) The students never measured the height directly, yet they can calculate it precisely. (2) Two students at different distances get different angles but should arrive at the SAME height — a powerful consistency check. (3) The same method, scaled up, is how surveyors map the Kenyan highlands, how engineers design the Thika Superhighway overpasses, and how foresters in the Mau Forest measure tree heights without felling them. The phenomenon is revisited and refined in every lesson: learners begin unable to explain it, gradually acquire the ratio tools (sin, cos, tan), discover the complementary-angle shortcut, build their own clinometer, and in the final lesson use the clinometer outdoors to measure a real structure and confirm their prediction — closing the explanatory loop completely.
Storyline Thread	Lesson 1 — PREDICT: "The Water Tank Problem" (Anchoring Phenomenon Launch) Learners observe a photograph/diagram of the school water tank or a similar elevated structure. In small groups they predict: "What information would you need, and how would you use it, to find the height of the tank without climbing?" Groups record predictions on sticky notes for the Driving Question Board (DQB). Teacher introduces vocabulary: right-angled triangle, hypotenuse, opposite side, adjacent side. Learners physically identify these sides in right-angled shapes around the classroom (door frames, window corners, book corners) with reference to a named acute angle. DQB is set up with the driving question posted prominently.

Lesson 2 — OBSERVE (Part 1): Deriving the Three Trigonometric Ratios

Learners use rulers and protractors to draw three different right-angled triangles each with the same acute angle (e.g., 35°) but different sizes. They measure the opposite, adjacent and hypotenuse for each triangle and compute the three ratios (opp/hyp, adj/hyp, opp/adj) in a table. Learners notice each ratio remains constant regardless of triangle size — a key discovery. Teacher names the ratios: sine, cosine, tangent. Learners use a scientific calculator to verify their measured ratios against calculator values. Connection to phenomenon: "These constant ratios are exactly what let us calculate the tank height from any distance!"

Lesson 3 — OBSERVE (Part 2): Sine and Cosine of Complementary Angles

Learners generate a table of angle pairs that sum to 90° (e.g., 20° & 70° , 35° & 55° , 45° & 45°) and record their sines and cosines. They discover that $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$. Learners construct an explanation for this pattern using a right-angled triangle diagram, noting that the "opposite" for one angle is the "adjacent" for its complement. DQB is updated: new understanding posted, new questions raised (e.g., "Does this work for ALL angles?").

Lesson 4 — EXPLAIN (Part 1): Applying Trigonometric Ratios to Find Unknown Sides

Guided practice: learners use sin, cos and tan to calculate unknown sides in right-angled triangles drawn on graph paper. Problems are set in Kenyan contexts — length of a roof rafter in Kisumu, distance across a stream in Kericho, height of a mango tree in a Mombasa compound. Learners practise selecting the correct ratio, setting up the equation and solving. Peer checking of answers. Connection to phenomenon: learners set up and solve the water-tank calculation for the first time using a given angle and distance.

Lesson 5 — OBSERVE (Part 3): Angle of Elevation — Clinometer Construction (Project Lesson)

Learners construct a simple clinometer using a cardboard protractor, drinking straw, string and a small weight (stone or metal washer). Teacher models how to sight along the straw to an elevated object and read the angle of elevation. Learners practise sighting objects of known approximate height in the classroom. Discussion: what is the "angle of elevation"? How does it relate to the right-angled triangle? DQB updated with new vocabulary and connections.

Lesson 6 — OBSERVE (Part 4): Angle of Depression and Outdoor Measurement

Learners take their clinometers outdoors. Working in groups of 3–4, they measure the angle of elevation to the top of the school flagpole, water tank, or a tall tree, and record the horizontal distance from the base. They also identify and measure an angle of depression (e.g., sighting from a raised veranda down to a ground-level marker). Data is recorded on structured worksheets. Back in the classroom, each group calculates the height of the object using the appropriate trigonometric ratio. Groups compare results and discuss sources of error.

Lesson 7 — EXPLAIN (Part 2): Problem Solving — Angles of Elevation and Depression

Structured problem-solving lesson using multi-step real-life scenarios: a Rift Valley farmer estimating the height of a cliff face, a diver at Watamu estimating depth from a boat, a construction worker in Nairobi checking the angle of a ramp. Learners work individually then compare with a partner. Common errors (confusing opposite/adjacent, not identifying the correct reference angle) are addressed explicitly. DQB is updated: learners move sticky notes from "questions" to "answered" and add new refined understandings. Learners revisit their original predictions from Lesson 1 and evaluate them.

Lesson 8 — MODEL BUILDING & SUMMATIVE CLOSURE: "Solving the Water Tank Problem"

Learners use all accumulated knowledge to produce a final written and diagrammatic solution to the original anchoring

	phenomenon. Each learner draws a labelled diagram of the water-tank scenario, identifies the right-angled triangle, selects and applies the correct trigonometric ratio, calculates the height, and writes a clear explanation of their method. Groups present their clinometer project findings (angle, distance, calculated height, reflection on accuracy). Teacher facilitates a whole-class DQB review — all original questions are revisited and either answered or identified as leading to future learning (e.g., trigonometry of non-right-angled triangles). Formative assessment via class activity and portfolio entry.
Supporting Phenomena	– THE LEANING SIGN BOARD: A rectangular signboard on a tilted pole outside the school gate — learners notice the pole makes an acute angle with the ground. How do we find the actual height the sign reaches without measuring along the pole? (Supports: identifying opposite/adjacent/hypotenuse and setting up a tan ratio.) – THE SHADOW PUZZLE: At midday, a vertical maize stalk in the school garden casts a short shadow; in the late afternoon the same stalk casts a much longer shadow. Why does the shadow length change, and can we use the shadow to find the height of the stalk? (Supports: angle of elevation of the sun, tan ratio, and the idea that the same height gives different ratios at different angles.) – THE MIRROR ON THE FLOOR: A small mirror is placed flat on the ground between a learner and the base of a wall. The learner steps back until they can just see the top of the wall in the mirror. The triangle formed is similar to the wall triangle. How is this related to the ratios we computed? (Supports: connecting similar triangles to constant trigonometric ratios.) – THE STAIRCASE RAMP: A newly built concrete ramp at a Nairobi hospital rises 1.2 m over a horizontal run of 4.8 m. Is this ramp accessible for wheelchair users (angle $< 14^\circ$)? (Supports: applying arctan/inverse trigonometry and angles of elevation in a real accessibility-design context.)

LESSON 1 (40 minutes): The Water Tank Problem — Launching the Anchoring Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon by activating prior knowledge of right-angled triangles and creating productive curiosity about how an unknown height can be determined using only an angle and a horizontal distance.
Knowledge	The learner identifies the sides of a right-angled triangle (opposite, adjacent, hypotenuse) in relation to a specific angle, and recognises that a triangle's shape is determined by its angles and at least one side length.
Skills	The learner makes and records a reasoned prediction about an unknown height; sketches and labels a right-angled triangle from a real-world scenario; identifies the hypotenuse, opposite side, and adjacent side relative to a given acute angle.
Attitudes	The learner demonstrates curiosity and openness by committing to a prediction before instruction and appreciates that mathematics can solve real-world problems — such as finding the height of a water tank — without direct measurement.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	This is the opening lesson of the sub-strand. It launches the anchoring phenomenon — 'How tall is the water tank without climbing it?' — and opens the Driving Question Board. Learners cannot yet answer the driving question; the lesson is designed to surface prior knowledge, generate genuine puzzlement, and establish the investigative thread that every subsequent lesson will build upon.
Safety Notes	No chemicals or sharp instruments are used in this lesson. If the class moves outdoors briefly to observe the school water tank, learners must remain on the footpath and not climb any structure. Teacher must conduct a quick headcount before and after any outdoor activity.

B. LESSON OVERVIEW

Lesson 1 launches the anchoring phenomenon by presenting learners with a puzzling, real-world challenge: determining the height of the elevated water tank on the school compound using only a measured horizontal distance and a sighted angle — no climbing, no tape measure along the height. The teacher uses a short dramatic scenario, a hand-drawn diagram on the board, and a structured Predict–Record–Share routine to surface what learners already know about right-angled triangles from Junior School. Learners make and justify individual predictions, sketch the scenario, and identify the triangle embedded in the problem. The lesson closes by opening the Driving Question Board (DQB), onto which learners post their initial questions and partial ideas. No trigonometric ratios are taught yet; the goal is productive puzzlement and a clear investigative purpose for the lessons ahead.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Volume word problem: water tank Source: Khan	Each learner receives or draws a blank diagram showing the school water tank on its steel frame, a student (Amina) standing 10 m from the base of the tower, and a dashed line from Amina's eye level	Before distributing diagrams, the teacher says: 'At many homesteads across Kenya — from the Rift Valley to the Coast — you see an elevated water tank on a steel frame. Our school has one	Predict-Record-Compare: Learners commit to a written prediction before any instruction, then immediately compare with peers. This strategy surfaces prior conceptions, creates cognitive	Observable indicator 1 — Diagram labelling: Teacher scans exercise books during circulation; target is that at least 70% of learners correctly identify and label a right-angled triangle (even informally as

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum) Search ARES for similar readings	to the top of the tank forming an angle with the horizontal ground. The diagram is unlabelled. Learners silently study the diagram for 2 minutes, then individually write answers to three prompts in their exercise books: (1) What shape do you see hidden in this picture? Label its sides. (2) Without calculating, estimate how tall you think the tank is — give a number in metres and explain your reasoning. (3) What information would you need to find the exact height? After writing, learners share predictions with a shoulder partner for 90 seconds, then selected learners share with the class. Predictions are recorded on the board in a two-column table: 'Our Estimate' and 'Our Reasoning'. The class observes that different groups gave different heights — all from looking at the same picture — creating the first puzzle.	too. Here is Amina, standing 10 metres from the base of the tower. She looks up at the top of the tank. I want you to think quietly about what you see.' Distribute or instruct learners to copy the diagram. Allow 2 minutes of silent thinking — teacher circulates, does NOT answer questions, notes 3–4 interesting sketches to reference later. After 2 minutes say: 'Write your three answers now. You have 3 minutes. Be specific — give a number for your height estimate.' After individual writing, say: 'Share your prediction with the person next to you. You have 90 seconds — GO.' After pair share, cold-call exactly 5 learners (choose gender balance and mix of confidence levels): 'Wanjiku, what height did you estimate and why?' — WAIT 12 seconds after each question before accepting an answer. Record all five estimates on the board without evaluating them. Then ask the whole class: 'Raise your hand if two people at your table gave the same estimate.' (Likely: very few.) 'Why do you think we all see the same picture but come up with different heights?' WAIT 15 seconds. Take 3 responses. Conclude: 'Today's big question is: can mathematics give us ONE exact answer — an answer we can all agree on — without anyone climbing that tank?'	dissonance when estimates differ, and gives every learner a personal stake in finding the correct answer. The visible range of estimates on the board makes the need for a precise mathematical method feel urgent and authentic.	'the slanted line, the ground, and the tower'). Observable indicator 2 — Prediction quality: Cold-called learners can state a specific number AND give at least one reason (e.g., 'I think about 8 metres because our classroom door is 2 metres and the tank looks about four doors high'). If fewer than half of five cold-called learners can give a reasoned estimate, the teacher revisits the diagram and asks 'What objects of known height can you compare the tank to?' before proceeding.
Observe Phase VIDEO: Trigonometry Word Problems Principles - Basic Source: CK-12 Search ARES for similar videos	The teacher reveals a second, fully labelled diagram on the board (or drawn large on manila paper) showing: (a) Amina standing 10 m from the base; angle of elevation = 38°; (b) Baraka standing 20 m from the same base; angle of elevation = 22°. Both sight lines go to the same point — the top of the	Reveal the labelled diagram and say: 'Here is the same water tank. Now I have given you real numbers. Two students — Amina and Baraka — both measure their angle of elevation to the top of the tank. Amina is 10 metres away and measures 38 degrees. Baraka is 20 metres away and measures	Anchored Observation with Structured Questions: Rather than a free-form 'what do you notice?', learners respond to two targeted written questions that direct attention to the key relationships (same height from different distances, inverse relationship between distance and angle). This	Observable indicator 1 — Diagram accuracy: Teacher checks that learners have correctly labelled the right angle, the two acute angles, and the three sides (opposite, adjacent, hypotenuse) relative to the angle of elevation. Learners who label 'hypotenuse' as the vertical side are flagged for

 EXERCISE: Ordering triangle sides and angles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	tank. The teacher marks the right angle at the base of the tower and labels the sides: 'horizontal distance (adjacent)', 'height of tank (opposite)', 'line of sight (hypotenuse)'. Learners copy the diagram into their books and answer two observation questions in writing: (1) Both Amina and Baraka are looking at the SAME point. Will they calculate the SAME height? Explain. (2) Amina is closer and her angle is bigger. Baraka is farther and his angle is smaller. Does this make sense? Why? Learners then share with their partner before a brief whole-class discussion. The teacher does NOT solve the problem — the point is to make the pattern (closer → steeper angle) visible and to introduce the vocabulary: angle of elevation, horizontal distance, opposite side, adjacent side, hypotenuse.	22 degrees.' Point explicitly to each measurement on the diagram. 'Copy this diagram carefully into your books — label every side and every angle I have shown.' Allow 3 minutes for copying. Then: 'Now answer my two questions in writing — 2 minutes.' After writing, cold-call 3 learners on Question 1 and 2 learners on Question 2 — WAIT 12 seconds each time. After responses, the teacher draws attention to the right angle: 'What is special about the angle at the base of the tower?' WAIT 10 seconds. Establish that it is 90°. Say: 'So we have a right-angled triangle. We know one side — the horizontal distance — and we know one angle. The question every lesson in this sub-strand will answer is: what is the rule, the ratio, the formula, that connects these pieces to the height?' Write the driving question on the board: 'How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment — without ever measuring it directly?' Tell learners: 'We are not going to answer that today. But you have just observed the evidence we will use in every lesson from now on.'	prevents surface-level observations and ensures every learner engages with the mathematical structure of the phenomenon before vocabulary is formalised.	targeted questioning: 'The hypotenuse is always opposite which angle?' Observable indicator 2 — Question 1 response: Learners should state that both students will get the same height (even if they cannot yet explain how). Learners who say 'different heights' are asked: 'If they are both looking at the same point on the tank, can the tank have two different heights at the same time?' This probes conceptual understanding of consistency.
Explain Phase  VIDEO: Volume word problem: water tank Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Trigonometry	Learners work in groups of four to complete a structured 'Triangle Anatomy' task on the diagram they copied. Each group receives (or draws) three versions of the same right-angled triangle at different scales but with the same angles (a set of similar triangles). For each triangle they measure: (a) the side opposite the marked angle, (b) the side adjacent to the marked angle, (c) the hypotenuse. They record	Say: 'In your groups, I want you to measure the three sides of each triangle on your sheet and fill in the table. Use your rulers — be precise to the nearest millimetre.' Allow 5 minutes for measurement and table completion, circulating to check ruler technique and that learners are correctly identifying opposite vs adjacent relative to the marked angle. After 5 minutes: 'Now divide opposite by adjacent	Ratio Invariance Discovery: By measuring physically and computing ratios across similar triangles of different sizes, learners encounter the core insight of trigonometry — that ratios of sides depend only on angles, not on size — before any formula is given. This builds conceptual readiness so that when sine, cosine, and tangent are formally introduced in Lesson 2, learners have an	Observable indicator 1 — Ratio computation: All four group members should have the same ratio written in their individual exercise books (not just the group recorder). Teacher checks 3–4 books per group during circulation. Observable indicator 2 — Explanation attempt: Spokesperson's sentence should reference the idea that the shape (angles) stays the same even

Word Problems: Helicopter Angle of Depression Rescue (PLIX) Source: CK-12 Search ARES for similar readings	measurements in a table. The teacher then asks groups to compute the ratio opposite ÷ adjacent for each triangle and record it. Groups notice the ratio is approximately the same across all three triangles despite different sizes — a striking result they cannot yet explain. Groups discuss: 'What does it mean that this ratio stays the same even when the triangle gets bigger?' Each group nominates a spokesperson to share one sentence summarising their finding. The teacher introduces the word 'trigonometry' (from Greek: trigonon = triangle, metron = measure) and explains that this sub-strand will be about using these consistent ratios to solve the water tank problem and problems like it across Kenya.	for each triangle. Use long division or your calculator if you have one. Write the answer to two decimal places.' Allow 2 minutes. Then cold-call one group: 'Fatuma, what ratio did your group get for the small triangle?' WAIT 12 seconds. Cold-call a second group for the medium triangle and a third for the large triangle. Write all three ratios on the board. Ask: 'These triangles are different sizes. Why is the ratio so similar?' WAIT 15 seconds — this is a genuine thinking pause, do not fill the silence. Take 3 responses. Do not confirm or deny explanations yet — say: 'Hold that thought. That is exactly what we are going to explain over the next several lessons. The word for this branch of mathematics is trigonometry.' Write 'TRIGONOMETRY' on the board and give the etymology. End by returning to the water tank diagram: 'If we can find the right ratio for a 38-degree angle, what can we calculate?' WAIT 10 seconds. Establish: the height of the tank — connecting back to the phenomenon.	experiential anchor for why those ratios are meaningful.	when the size changes. Sentences that only describe the arithmetic ('we divided and got the same number') without referencing the triangle prompt a follow-up: 'Why do you think the shape of the triangle might have something to do with the ratio staying constant?'
Driving Question Board (DQB) Creation  VIDEO: Volume word problem: water tank Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Right triangle trigonometry	Each learner is given two small slips of paper (or writes in a designated DQB section of their book if slips are not available). On slip 1 (yellow / marked Y): they write one question the lesson has raised for them that they do not yet know the answer to. On slip 2 (blue / marked B): they write one thing they are now more confident about compared to the start of the lesson. Learners post / record their slips. The teacher reads five questions aloud and organises them into two clusters on the board: (A) Questions about HOW	Distribute slips or instruct learners to draw two boxes in their books. Say: 'You have 2 minutes to write — one genuine question on your yellow slip, one thing you are more confident about on your blue slip. Do this individually and in silence.' After 2 minutes, collect or have learners display slips. Read five questions aloud — choose a mix of procedural ('How do I find the height?') and conceptual ('Why does the ratio stay the same for all similar triangles?'). Organise visibly on the board. Say: 'These are YOUR questions. They are not	Question Sorting on the DQB: Publicly organising learner questions into HOW and WHY clusters gives learners metacognitive visibility into the structure of the sub-strand's inquiry. It also communicates that procedural questions (how to calculate) and conceptual questions (why the method works) are both valued — preventing the common misconception that mathematics is only about procedures.	Observable indicator 1 — Question quality: At least 80% of learners produce a question that is specific to the phenomenon (e.g., 'How do we use 38 degrees to find the height?') rather than generic (e.g., 'What is trigonometry?'). Generic questions prompt teacher follow-up: 'Can you connect your question to Amina, Baraka, and the water tank?' Observable indicator 2 — Confidence statement: Learners should articulate something concrete (e.g., 'I am more confident that a right-angled triangle is hidden in

word problems Source: Khan Academy (English - US curriculum) Search ARES for similar readings	(e.g., 'How do we actually calculate the height from the angle?') and (B) Questions about WHY (e.g., 'Why does the ratio stay the same?'). The DQB now displays the driving question at the top, with learner questions beneath it. The teacher tells learners: 'We will answer every one of these questions before the end of this sub-strand. Each lesson, we will come back to this board and move questions from the unanswered side to the answered side.'	silly — they are exactly the questions that Kenyan surveyors, engineers on the Thika Superhighway, and foresters in the Mau Forest needed to answer. By the end of this sub-strand, you will be able to answer all of them.' Point to the driving question at the top of the DQB. 'This is our compass. Every lesson, we come back here.' Leave the DQB visible for all subsequent lessons.		the water tank problem') rather than vague ('I know more now'). Vague statements are revisited at the start of Lesson 2.
Model Building Phase  VIDEO: Volume word problem: water tank Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners draw their first 'Explanatory Model' — a labelled sketch in a dedicated 'Model' page at the back of their exercise book (or on a card that will be kept across all lessons in the sub-strand). The model for Lesson 1 must include: (1) The water tank scenario with Amina's position, the 10 m horizontal distance, the 38° angle of elevation, and the unknown height marked as 'h = ?'. (2) The right-angled triangle isolated from the scenario, with all three sides labelled using the words: OPPOSITE (height), ADJACENT (horizontal distance), HYPOTENUSE (line of sight). (3) A note in their own words: 'The ratio opposite ÷ adjacent seems to stay the same for triangles with the same angles.' Learners annotate their model with one arrow pointing to the unknown 'h' and the label: 'This is what trigonometry will help us find.' The model is explicitly incomplete — learners write 'MODEL VERSION 1 — TO BE UPDATED' at the top.	Say: 'Every scientist and engineer keeps a model — a working diagram that shows what they currently understand and what they still need to figure out. You are going to keep a trigonometry model across all our lessons. Open to the last page of your exercise book — or use the card I am giving you. This is your Model Page.' Allow 5 minutes for drawing and labelling. Circulate and check three things: (1) Is 'h = ?' marked on the vertical side? (2) Are all three side-names written correctly relative to the 38° angle? (3) Is the ratio note written in their own words — not copied verbatim from the board? If a learner has the hypotenuse incorrectly placed, kneel to their level and ask quietly: 'In a right-angled triangle, which side is always opposite the right angle?' WAIT 10 seconds. After 5 minutes, say: 'Write at the top: MODEL VERSION 1 — TO BE UPDATED. In our next lesson, you will add the first trigonometric ratio to this model. In our final lesson, you will go outside with a clinometer you have built yourself and measure the actual height of	Progressive Model Revision: Establishing a persistent, explicitly versioned model from Lesson 1 creates a tangible record of growing understanding across the sub-strand. Marking the model as incomplete ('TO BE UPDATED') normalises scientific revision and prevents learners from treating each lesson as an isolated topic. Connecting the model to the final outdoor measurement in the closing lesson gives the model a purpose beyond the current lesson.	Observable indicator 1 — Correct triangle anatomy: Teacher checks that opposite, adjacent, and hypotenuse are labelled relative to the 38° angle of elevation (NOT the right angle or the unlabelled acute angle). At least 75% of models should be correctly labelled by end of lesson. Observable indicator 2 — Own-words ratio note: Learner's written note should paraphrase the invariance observation in their own language. Direct copying of the board statement without personalisation is flagged: teacher asks learner to close their book and explain the idea aloud, then rewrite.

		our water tank — and check whether your model predicted it correctly.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following four questions and record brief notes:

- (1) PREDICTION RANGE — How wide was the spread of height estimates from the class? A very narrow spread (all estimates within 1 m of each other) may suggest learners already have strong spatial intuition, or that they converged socially rather than reasoning independently — adjust Lesson 2 to probe individual reasoning more carefully. A very wide spread (estimates from 2 m to 30 m) confirms productive uncertainty and validates launching the sub-strand from this phenomenon.
- (2) TRIANGLE ANATOMY ACCURACY — What proportion of learners correctly labelled opposite, adjacent, and hypotenuse relative to the angle of elevation (NOT the right angle) in Phase 3 and Phase 5? If fewer than 60% were correct, begin Lesson 2 with a 5-minute retrieval activity explicitly distinguishing 'side opposite the right angle' from 'side opposite a specific acute angle'.
- (3) DQB QUESTION QUALITY — Were learner questions on the DQB specific enough to drive inquiry? If most questions were generic ('What is $\sin$?'), plan a 'Question Refinement' warm-up in Lesson 2 where learners rewrite their DQB question to include the water tank context.
- (4) ENGAGEMENT AND EQUITY — Which learners did NOT contribute during cold-calls or pair-share? Note names. In Lesson 2, deliberately cold-call at least two of these learners on lower-stakes questions (e.g., 'Point to the hypotenuse in your diagram') to build their confidence before moving to more demanding questions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Two students (Amina at 10 m with a 38° angle of elevation, and Baraka at 20 m with a 22° angle of elevation) both sight the top of the same school water tank from different positions on the ground. Despite being at different distances and measuring different angles, they should calculate the same height — because the tank has only one height. We also observed that when we measured the ratio of the opposite side to the adjacent side across three right-angled triangles of different sizes but the same angles, the ratio stayed nearly the same.
What did I learn?	A right-angled triangle is hidden inside the water tank problem: the vertical height of the tank is the OPPOSITE side, the horizontal distance from the observer to the base is the ADJACENT side, and the line of sight to the top of the tank is the HYPOTENUSE — all defined relative to the angle of elevation. The ratio of two sides of a right-angled triangle appears to depend only on the angle, not on the size of the triangle. The branch of mathematics that studies these ratios is called TRIGONOMETRY.
How does this explain the phenomenon?	We cannot yet fully explain WHY the ratio stays constant, or HOW to use the 38° angle and 10 m distance to calculate the exact height of the tank. These questions are now posted on our Driving Question Board. We also cannot yet explain why a larger distance produces a smaller angle of elevation to the same point — we have observed it but need the trigonometric ratios (introduced in Lesson 2) to make sense of it mathematically.

LESSON 2 (40 minutes): Sides of the Right-Angled Triangle — Naming What We See

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners establish precise vocabulary for the three sides of a right-angled triangle — hypotenuse, opposite, and adjacent — always defined relative to a named acute angle, building the conceptual language needed to express trigonometric ratios in subsequent lessons.
Knowledge	– Identify and name the three sides of a right-angled triangle (hypotenuse, opposite side, adjacent side) relative to a specified acute angle. – Recognise that the labels 'opposite' and 'adjacent' switch when the reference angle changes. – Locate right-angled triangles embedded in real structures (door frames, window corners, the water-tank diagram).
Skills	– Correctly label all three sides of any right-angled triangle given a named acute angle. – Re-label the sides accurately when the reference angle shifts from one acute angle to the other.
Attitudes	– Appreciate that precise mathematical language is a tool that makes communication exact and avoids errors. – Show curiosity about how the same triangle can be described differently depending on the chosen reference angle.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	In Lesson 1, learners discovered that a right-angled triangle is hidden inside the water-tank problem and that the ratio of two sides seems to depend only on the angle. Before those ratios can be calculated or named, learners need shared, unambiguous labels for the sides. This lesson gives them exactly that vocabulary — hypotenuse, opposite, adjacent — so that in Lesson 3 they can define sine, cosine and tangent as specific ratios of these labelled sides.
Safety Notes	No specific hazards. If learners move around the classroom to measure door frames or window corners, remind them to move calmly and not obstruct walkways.

B. LESSON OVERVIEW

Lesson 2 is a vocabulary-and-concept lesson that sits at the bridge between the anchoring phenomenon (the water-tank problem) and the formal definition of trigonometric ratios. Learners arrive already knowing — from Lesson 1 — that a right-angled triangle is embedded in the water-tank scenario: the vertical height of the tank is one side, the horizontal ground distance is another, and the line of sight is the longest side. What they lack is a shared language for these sides that will allow the teacher and textbook to write down ratios unambiguously. This lesson fills that gap by having learners physically handle right-angled triangles cut from paper, identify the right angle, choose a reference acute angle, and then name each side according to its spatial relationship to that angle.

The central cognitive move of the lesson is the discovery that the labels 'opposite' and 'adjacent' are not fixed properties of a side — they depend entirely on which acute angle the learner is currently looking at. When the reference angle shifts, the opposite and adjacent sides swap. Only the hypotenuse (the side opposite the right angle, always the longest side) keeps its name regardless of which acute angle is chosen. This is a powerful and slightly surprising result that learners encounter through hands-on re-labelling activities before any formal instruction, ensuring the idea is constructed rather than merely received.

Throughout the lesson, every labelling exercise is tied back to the water-tank diagram: the teacher consistently asks 'Which side is this in our water-tank triangle?' so that abstract vocabulary is always grounded in the anchoring phenomenon. By the end of the lesson, every learner can look at the water-tank sketch on the board, point to the angle of elevation, and correctly name all three sides. This positions the class perfectly for Lesson 3, where the ratios opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent will be given their trigonometric names.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner receives a small pre-cut right-angled triangle (cut from plain paper or a sheet of graph paper — prepared by the teacher before class). The teacher marks one acute angle with the letter θ (theta) using a marker. Working individually, learners study the triangle and write on their exercise book or sticky note: (1) Which side do they think is the most important or special, and why? (2) If they had to give each side a unique name that describes its relationship to the marked angle θ, what names would they choose? Learners write their proposed names without any prompting from the teacher. After 3 minutes, partners share their proposed names with each other.	Distribute pre-cut triangles and markers before the lesson begins to save time. Display the water-tank diagram from Lesson 1 on the board (drawn or projected) with the angle of elevation marked as θ. Say: 'Last lesson we agreed there is a right-angled triangle hiding inside the water-tank problem. Today I want to know — before I teach you anything — what name would YOU give to each of these three sides?' Give WAIT TIME of 15 seconds after posing the question to allow private thinking before writing. Circulate quietly; do NOT correct or affirm any proposed names at this stage. After 3 minutes, cold-call 4 learners: 'Akinyi, what name did you give to the longest side and why?' — 'Otieno, did you give the same name or a different one?' — 'Wanjiku, which side did you find hardest to name?' — 'Mwangi, does the angle θ affect your naming at all?' Record the variety of proposed names on the board in a 'Learner Ideas' column without editing them. Say: 'These are all interesting ideas. Let us now look at some evidence to see which naming system mathematicians settled on — and why.'	Elicit-Before-Instruction (EBI): By asking learners to generate their own names before formal vocabulary is introduced, the teacher activates prior spatial knowledge, surfaces informal language (e.g. 'the slanted one', 'the tall side', 'the bottom side'), and creates a personal investment in discovering the 'official' terms. The subsequent reveal feels like resolving a genuine question rather than receiving an arbitrary definition.	Teacher scans written predictions and listens to partner-share conversations. Key indicator: Do learners mention the right angle or the marked angle θ when justifying their names? Learners who ignore the angle θ entirely will need extra support during the Observe Phase to grasp that 'opposite' and 'adjacent' are angle-relative labels. Teacher makes a mental note of these learners for targeted cold-calling in the Explain Phase.
Observe Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES	The teacher draws (or projects) a clear right-angled triangle on the board — labelled with vertices A (right angle), B, and C — and marks angle θ at vertex B. The teacher then introduces the three side names by pointing and narrating, using the water-tank diagram alongside the abstract	Draw a large, clear right-angled triangle on the board with vertices labelled A (right angle, marked with a small square), B, and C. Mark angle θ at vertex B. Point to side AC (the side directly opposite θ) and say: 'This side is directly across from our angle θ — it faces θ. Mathematicians call it the	Concrete-Representational-Abstract (CRA) progression: Learners move from handling a physical paper triangle (concrete), to labelling drawn diagrams (representational), to identifying sides in real classroom objects (abstract application). Tying each step back to the water-tank	During the scavenger activity, teacher listens for correct use of all three terms. Success indicator: learner names the hypotenuse correctly regardless of which acute angle is chosen (hypotenuse never changes) AND correctly swaps opposite/adjacent when the reference angle moves. Common

for similar videos  HTML: Right Triangle Trigonometry (Flexbook) Source: CK-12  Search ARES for similar readings	triangle. Learners observe carefully and copy the labelled diagram into their exercise books. Then, in pairs, learners are given a second activity: on a printed or hand-drawn worksheet (or directly in their books), they are shown the same triangle but this time angle θ is moved to vertex C. Pairs re-label all three sides for this new reference angle. Next, learners stand up and identify one real right-angled shape in the classroom (door frame, window corner, corner of a textbook, corner of a desk) — they physically point to and name the hypotenuse, opposite side, and adjacent side relative to an imaginary angle at one corner, then report back to their partner.	OPPOSITE side.' Write 'Opposite' along that side. Point to side AB (the side running from θ to the right angle) and say: 'This side is right next to our angle θ — it is touching θ but is not the hypotenuse. It is called the ADJACENT side — adjacent means neighbouring.' Write 'Adjacent' along that side. Point to side BC (the longest side, opposite the right angle) and say: 'This side is opposite the right angle itself — it is always the longest side. It is called the HYPOTENUSE.' Write 'Hypotenuse' along that side. Then say: 'Now watch what happens if I move the reference angle.' Erase the θ label and redraw it at vertex C. Give WAIT TIME of 12 seconds and ask: 'Has anything changed about the hypotenuse?' Cold-call 2 learners. Then ask: 'What happened to the opposite and adjacent sides?' Cold-call 3 learners. Explicitly connect to the water-tank: 'In our water-tank triangle — the vertical height of the tank is the ____? The ground distance is the ____? The line of sight is the ____?' Cold-call 3 different learners. For the classroom scavenger activity, say: 'You have 3 minutes. Find one right angle in this room, decide on your reference angle, and name all three sides out loud to your partner.' Circulate and listen. After 3 minutes, call on 3 pairs to share: 'Tell us your right angle, your reference angle, and your three side names.'	diagram ensures that the abstract vocabulary is always anchored in the phenomenon, preventing rote memorisation of disconnected terms.	error to watch for: learners who label 'adjacent' as the horizontal side and 'opposite' as the vertical side regardless of the reference angle — this indicates they are using positional (up/down/across) logic rather than angle-relative logic. Flag these learners for additional questioning in the Explain Phase.
Explain Phase  VIDEO: Solving for a side in right triangles with trigonometry	Learners work in groups of three or four on a structured sensemaking task. Each group receives a worksheet (or copies from the board) showing three	Before groups begin, say: 'I want you to use the words hypotenuse, opposite, and adjacent in every answer — not the words tall, slanted, or bottom.' Post the three	Structured Group Sensemaking with Public Representation: Groups construct meaning collaboratively and then make their thinking visible to the whole class	Teacher evaluates each group's written labelling for three specific indicators: (1) Hypotenuse correctly identified in all three triangles regardless of θ position.

Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Solve for a side in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	different right-angled triangles, each with a different acute angle marked as θ. For each triangle, groups must: (a) identify and label all three sides; (b) write one sentence explaining why a particular side is the hypotenuse; (c) write one sentence explaining why the 'opposite' label would move if θ moved to the other acute angle. Groups then answer the key connection question: 'In the water-tank diagram, the angle of elevation is θ. Write the name of each side of the water-tank triangle using today's vocabulary.' Each group writes their answer on a large sheet of paper or on the board section assigned to them. A representative from each group presents their answer to the class in 30 seconds.	vocabulary words on the board with brief definitions as a reference card. While groups work (allow 7 minutes), circulate and use targeted prompts: For groups using positional language, ask: 'If I tilt this triangle 45 degrees on the page, does the hypotenuse change? Does the opposite side change? What does that tell you about how the names work?' For groups who finish early, challenge: 'What if I do not mark any angle at all — can I still identify the hypotenuse? Can I identify the opposite side?' During presentations, after each group presents, give WAIT TIME of 10 seconds before asking the class: 'Does anyone want to agree, disagree, or add to what this group said?' Cold-call 2 learners per group presentation (10 cold-calls total across 4–5 groups). Conclude with a whole-class synthesis: draw the water-tank triangle on the board and call on 3 cold-called learners to label the three sides one at a time: 'Chebet — come and write the name of the longest side.' 'Kamau — label the side opposite the angle of elevation.' 'Atieno — label the remaining side.' Affirm correct answers and correct errors with explanation.	by writing on board sections or large paper. The public display allows learners to compare multiple representations simultaneously, notice agreements and discrepancies, and build shared understanding. Presenting in 30-second summaries forces groups to distil their thinking to the essential claim rather than reciting every step.	(2) Opposite and adjacent correctly swapped when θ moves. (3) Water-tank triangle correctly labelled with the angle of elevation as the reference angle. Any group that fails indicator (2) is flagged — teacher re-teaches the angle-relative nature of opposite/adjacent using a physical demonstration (rotating a paper triangle) before moving to the DQB phase.
Driving Question Board (DQB) Creation  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners individually write one sticky note (or a folded card) responding to the prompt: 'We now know the names of the three sides of the water-tank triangle. What NEW question does this make you want to ask about how to actually find the height of the tank?' Learners post their sticky notes on the class Driving Question Board under the driving question: 'How can we use the angles and one	Distribute sticky notes or ask learners to tear a small piece of paper from their exercise books. Say: 'Take 2 minutes. Write one question that today's vocabulary makes you curious about — specifically a question about the water-tank problem.' Give full WAIT TIME of 2 minutes of silent individual writing. Do NOT prompt or suggest questions during this time. After learners post their	Driving Question Board as Collective Sensemaking Tool: The DQB functions as a public, cumulative record of the class's growing understanding and evolving questions. Posting new questions helps learners see that understanding a concept does not close the inquiry — it opens new, more precise questions. This positions the DQB as a living	Teacher reads all posted questions before the next lesson. Success indicator: at least 60% of questions reference the angle, a ratio, or a calculation (e.g. 'How does the angle tell us the length of the opposite side?'). This shows learners are connecting the new vocabulary to the phenomenon's core puzzle. Questions that only re-ask Lesson 1 content (e.g. 'What is a right angle?') indicate

 EXERCISE: Solve for an angle in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	side of a right-angled triangle to find any unknown height or distance in our environment — without ever measuring it directly?' The teacher reads several questions aloud, groups similar questions together, and labels clusters. Learners also update the 'What We Now Know' column of the DQB with today's key learning: 'The three sides of a right-angled triangle are named hypotenuse, opposite, and adjacent — relative to a chosen acute angle.'	notes, read 5–6 questions aloud — choose a mix of simple and sophisticated questions. Say: 'I notice several of you are asking about the relationship between the angle and the sides — for example, Njeri asks: does a bigger angle of elevation always mean a bigger opposite side? That is exactly the question Lesson 3 will answer.' Point to the driving question on the DQB: 'Keep our big question in mind: we need the angle AND a side length to find the tank height. We now know what to call the sides. Next, we need to discover the mathematical relationship between the angle and the ratio of those sides.' Add a new 'anchor card' to the DQB: a small diagram of the water-tank triangle with all three sides labelled using today's vocabulary. Cold-call 2 learners: 'Mutua — looking at this diagram, which side of the water-tank triangle is opposite the angle of elevation?' 'Auma — which side is the hypotenuse?'	document that mirrors the actual practice of mathematical inquiry.	those learners need vocabulary consolidation before Lesson 3.
Model Building Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Use Pythagorean theorem to find right triangle side lengths Source: Khan Academy (English	Learners return to the water-tank diagram they sketched in Lesson 1 (or redraw it if needed). Using the vocabulary from today, they annotate their diagram with the labels: angle of elevation (θ), hypotenuse (line of sight), opposite side (vertical height of the tank — the unknown we want to find), and adjacent side (horizontal ground distance — the known measurement). Below the diagram, learners write a two-sentence 'model statement' in their own words: (1) what information is known in the water-tank problem, and (2) what information is unknown and needs to be calculated. This annotated	Say: 'Open your exercise books to the water-tank sketch from Lesson 1. If you do not have it, draw a fresh one — a right-angled triangle with a small person at the bottom-left, the tank height as the vertical side, and the ground distance as the horizontal side.' Give 2 minutes for drawing. Then say: 'Using today's vocabulary, label every part of your triangle. Do not leave any side or angle unnamed.' Give WAIT TIME of 3 minutes for labelling. Circulate and check: every diagram must show θ at the observer's position, 'Hypotenuse' along the line of sight, 'Opposite' along the vertical height, and 'Adjacent' along the ground.	Cumulative Model Annotation: Rather than building a new model each lesson, learners continuously annotate and refine the same water-tank diagram. This strategy makes the growth of understanding physically visible — each new label, arrow, or equation added to the diagram represents a new conceptual tool acquired. It also reinforces the storyline by keeping the phenomenon central to every lesson's output.	Teacher collects or spot-checks 6–8 exercise books at the end of the lesson. Success criteria: (1) All three sides labelled correctly with the new vocabulary. (2) θ placed at the observer's acute angle — not at the right angle. (3) Two-sentence model statement correctly identifies 'adjacent' (ground distance) as known and 'opposite' (tank height) as unknown. Learners who place θ at the wrong angle or who label the hypotenuse as 'opposite' need individual follow-up at the start of Lesson 3.

- US curriculum) Search ARES for similar readings	diagram is kept in their exercise books and will be updated in every subsequent lesson as new tools are added.	Correct any errors quietly by asking guiding questions: 'Which side faces directly across from the angle θ?' After labelling, say: 'Now write two sentences below your diagram. Sentence 1: What do we know in this problem? Sentence 2: What are we trying to find — and what is its label?' Cold-call 3 learners to read their two sentences aloud. Affirm and display one strong example on the board. Close with: 'Your diagram now has a complete language. In Lesson 3, we will discover the mathematical ratio that connects the angle θ to the opposite and adjacent sides — and that ratio will give us the height of the water tank.'		
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D. TEACHER REFLECTION

1. When learners re-labelled the triangle after moving θ from one acute angle to the other, did the majority correctly swap 'opposite' and 'adjacent' while keeping 'hypotenuse' fixed? If not, what classroom evidence suggested they were using positional rather than angle-relative logic?

2. How effectively did the classroom scavenger activity (finding real right-angled shapes) help learners generalise the vocabulary beyond drawn diagrams? Were there any learners who could label diagrams correctly but struggled when the triangle was embedded in a physical object?

3. Review the DQB sticky notes: what proportion of learners' new questions explicitly mention the angle, a ratio, or a calculation? Does this suggest the class is ready for the formal definition of trigonometric ratios in Lesson 3, or is consolidation of today's vocabulary needed first?

4. The water-tank diagram annotation is the core cumulative model for this unit. Were learners' two-sentence model statements precise enough to distinguish between 'known' (adjacent/ground distance) and 'unknown' (opposite/tank height)? What language scaffolding might help weaker writers in future lessons?

5. Did the lesson maintain a consistent connection to the anchoring phenomenon throughout all five phases? At which phase did the water-tank connection feel weakest, and how could that phase be adjusted in the next teaching cycle?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that in a right-angled triangle, the three sides have specific names — hypotenuse, opposite, and adjacent — and that the labels 'opposite' and 'adjacent' depend on which acute angle is chosen as the reference angle. They observed this by labelling paper triangles, re-labelling when the reference angle moved, and identifying right-angled shapes in real classroom objects including the water-tank diagram.
What did I learn?	The hypotenuse is always the side opposite the right angle and is always the longest side — its name never changes. The 'opposite' side is the side directly facing the chosen acute angle θ, and the 'adjacent' side is the side next to θ that is not the hypotenuse. When θ moves to the other acute angle, opposite and adjacent swap. In the water-tank triangle: the line of sight = hypotenuse, the tank height = opposite, the ground distance = adjacent.

How does this explain the phenomenon?

We now have a precise vocabulary to describe the water-tank triangle. The height of the tank that we cannot measure directly is the 'opposite' side, and the ground distance we can measure is the 'adjacent' side. In future lessons, we will use the ratio of these two sides — together with the angle of elevation — to calculate the exact height of the tank without climbing it.

LESSON 3 (80 minutes (double lesson)): Deriving the Three Trigonometric Ratios: Sine, Cosine and Tangent

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners derive and apply the three trigonometric ratios — sine, cosine, and tangent — from right-angled triangles, directly connecting ratio values to the water-tank measurement challenge.
Knowledge	The learner identifies the opposite side, adjacent side, and hypotenuse of a right-angled triangle relative to a specific acute angle, and states the definitions of $\sin \theta$, $\cos \theta$, and $\tan \theta$ as ratios of these sides.
Skills	The learner measures sides of drawn right-angled triangles, computes the three ratios for given angles, uses a calculator or mathematical tables to find ratio values, and applies $\tan \theta$ to calculate an unknown height given a horizontal distance and angle of elevation.
Attitudes	The learner appreciates that fixed angle–ratio relationships make indirect measurement precise and reliable, and demonstrates patience and care when measuring and recording triangle dimensions.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	In Lesson 2 learners confirmed through paper-folding and measurement that similar right-angled triangles share constant side ratios for the same angle. In this lesson they NAME those ratios ($\sin$, $\cos$, $\tan$), practise computing them, and make the first direct calculation of the water-tank height — showing that $\tan \theta$ alone, combined with a measured horizontal distance and a sighted angle, is sufficient to find any inaccessible vertical height.
Safety Notes	Learners use rulers, protractors, and scissors with care. Sharp points should be directed away from peers. No climbing of furniture or structures is permitted during any outdoor sighting activity in this lesson.

B. LESSON OVERVIEW

Learners begin by revisiting their Lesson 2 finding — that the ratio opposite/adjacent stays constant for a fixed angle regardless of triangle size — and are told this ratio has a name: tangent. Through a structured draw-and-measure activity using three right-angled triangles with angles of 30° , 45° , and 60° , learners derive all three ratios ($\sin$, $\cos$, $\tan$) for each angle, record them in a class ratio table, and verify the values with a calculator. They then apply $\tan \theta$ directly to the anchoring phenomenon, calculating the height of the school water tank from two different horizontal distances and confirming that both give the same height — exactly the consistency check posed at the start of the unit. The lesson closes with learners updating their Driving Question Board and revising their explanatory models to include ratio notation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Finding reciprocal trig ratios Source: Khan Academy (English)	Learners are shown a projected diagram of the school water-tank scenario: a student standing 10 m from the base of the tank tower sights the top at an angle of elevation of 38° . A second student	Display the water-tank diagram on the board or printed card. Say: 'Look at this carefully. You have an angle and a horizontal distance. In your book, predict how you would use those two pieces of	Prediction journaling with annotated sketches. This activates prior knowledge from Lesson 2 (constant ratios in similar triangles) and creates cognitive dissonance — learners who predict different	Circulate and scan exercise books during the 3-minute silent prediction. Look for: (1) learners who draw a right-angled triangle and label angle/sides correctly — these learners are ready to extend;

- US curriculum) Search ARES for similar videos EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	stands 15 m away and sights the top at 27°. The teacher asks: 'Both students are trying to find the same height. Will they get the same answer? What information connects the angle and the distance to the height?' Learners write a private prediction (2–3 sentences) in their exercise books, including a sketch. They then share with a shoulder partner for 90 seconds before the teacher cold-calls 4 pairs to share predictions with the class.	information to find the height. Draw a sketch to support your prediction. You have 3 minutes — work silently first.' After 3 minutes, say: 'Now share your prediction with your partner for 90 seconds.' After sharing, cold-call 4 pairs using name cards: 'Amina and Juma — what did you predict, and why?' After each response, press with: 'What ratio or operation are you imagining?' Record key prediction words on the board (e.g., 'divide', 'multiply', 'the angle tells us the steepness'). Do NOT correct misconceptions yet — annotate them as 'to be tested'.	answers for different distances will be surprised when both yield the same height, motivating careful attention during the Observe phase.	(2) learners who write 'I add the angle to the distance' or do not connect angle to ratio — flag for targeted questioning during Phase 2. Collect a mental tally: roughly how many learners have spontaneously written a ratio (opposite over adjacent) versus those who have not.
Observe Phase VIDEO: Intro to the trigonometric ratios Source: Khan Academy (Kiswahili) Search ARES for similar videos EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	In groups of three, learners receive a printed activity sheet with three pre-drawn right-angled triangles, each with a different marked acute angle (30°, 45°, 60°) and one side labelled. Step 1 — Label: Learners identify and label the hypotenuse, the side opposite the marked angle, and the side adjacent to the marked angle for each triangle. Step 2 — Measure: Using a ruler, learners measure all three sides (in mm) and record values in a table. Step 3 — Compute: Learners calculate opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent for each triangle, rounding to 2 decimal places. Step 4 — Verify: Learners enter the angle into a calculator (sin, cos, tan keys) and compare the displayed value with their computed ratio. Step 5 — Observe the pattern: Learners record what they notice about the ratio values across different triangle sizes drawn by different groups but sharing the same angle.	Distribute printed activity sheets and rulers. Say: 'Your job in the next 20 minutes is to measure, calculate, and discover the names for three ratios that live inside every right-angled triangle. Start with Step 1 — label your sides. Remember from last lesson: the hypotenuse is always opposite the right angle. The opposite and adjacent sides change depending on WHICH angle you choose.' Circulate after 4 minutes. Crouch to group level; ask: 'Which side is opposite the 30° angle in your triangle — point to it.' If a group is confused, ask: 'Where is the right angle? Good. Where is the 30° angle? Now the side across from 30° — that is the opposite side.' After Step 3, pause the class: 'Pens down, eyes up. I want to hear ratio values. What did Group 1 get for opposite over hypotenuse for the 45° triangle?' Cold-call 3 groups. Write values on the board. Say: 'Interesting — they are very close. Why might there be small differences?' (Measurement error.) 'Now check with the sin key on	Measure–compute–verify cycle (Data-driven ratio discovery). By physically measuring triangle sides and computing ratios before pressing the calculator key, learners construct the meaning of sin, cos, and tan from concrete evidence rather than receiving them as abstract definitions. Comparing computed values across groups demonstrates the constancy of ratios — the central insight needed to solve the phenomenon.	During Step 1 labelling, scan for correct identification of opposite vs adjacent (common error: swapping opposite and adjacent when the angle is at the bottom-left vs bottom-right of the triangle). During Step 3, listen for learners computing hypotenuse/opposite rather than opposite/hypotenuse — redirect immediately: 'Which is on top in your fraction — the smaller side or the hypotenuse?' After Step 4 calculator verification, ask the class: 'Thumbs up if your computed value matched the calculator to 1 decimal place.' A show of fewer than 70% thumbs signals a need to address measurement precision before proceeding.

		your calculator. What does it say for $\sin 45^\circ$?' Pause 10 seconds. Cold-call 2 learners: 'Fatuma — what did your calculator show?' Introduce formal names: 'We give these three ratios special names. Opposite over hypotenuse is the SINE of the angle — written $\sin \theta$. Adjacent over hypotenuse is the COSINE — $\cos \theta$. Opposite over adjacent is the TANGENT — $\tan \theta$.' Write on the board in large text: $\sin \theta = O/H$, $\cos \theta = A/H$, $\tan \theta = O/A$. Say the mnemonic slowly: 'SOH — CAH — TOA. Repeat it with me.' Drill the mnemonic twice as a class.		
Explain Phase  VIDEO: The trig functions & right triangle trig ratios Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners apply the tangent ratio to the water-tank problem. The teacher works the first calculation as a shared example (Student A: 10 m from base, angle of elevation 38°). Learners then independently calculate the height using Student B's data (15 m from base, angle of elevation 27°). After completing both calculations, each learner writes a 2-sentence explanation of why the two answers should be the same height and what this tells us about trigonometric ratios. Groups then compare answers and discuss any discrepancies, attributing differences to rounding or measurement error rather than a flaw in the method.	Say: 'Now let us return to the water tank. Student A stands 10 metres from the base and sights the top at 38°. I want to find the height h. Let me draw this on the board.' Draw a right-angled triangle: horizontal base = 10 m (adjacent), angle at A = 38°, vertical side = h (opposite). Say: 'Which ratio connects the opposite side and the adjacent side?' Wait 10 seconds. Cold-call: 'Kipchoge — which ratio?' (Expected: tangent.) 'Correct. So $\tan 38^\circ = h/10$. Rearranging: $h = 10 \times \tan 38^\circ$. Everyone find $\tan 38^\circ$ on your calculator.' Wait 15 seconds. 'What did you get?' Cold-call 3 learners. Write: '$\tan 38^\circ \approx 0.7813$. So $h \approx 10 \times 0.7813 = 7.81$ m.' Say: 'Now YOU do Student B: 15 metres away, angle 27°. Set up the equation, solve it, and write your answer. Silently. 5 minutes.' Circulate and observe written work — do not help unless a learner is completely stuck; prompt with 'Which ratio?' only. After 5 minutes, cold-call 3 learners to share their answer. Write the correct answer on the board ($h \approx$	Worked-example transfer (I do → You do). The teacher models the algebraic setup explicitly, then immediately releases responsibility to learners for a parallel problem. The two-student consistency check is the explanatory payoff of the entire anchoring phenomenon: learners viscerally experience why the method works. The Kenya-context extension questions connect the abstract ratio to professional practice.	Collect all exercise books after the independent calculation. Scan for three specific errors: (1) learner writes $\tan 27^\circ = 15/h$ instead of $h/15$ (ratio inverted — common); (2) learner uses $\sin$ or $\cos$ instead of $\tan$ (wrong ratio selected); (3) learner sets up correctly but makes an arithmetic error. Tally error types on a sticky note for use in Lesson 4 warm-up remediation. Any learner who completes the calculation correctly AND gives a coherent written explanation is performing at 'Meets Expectations' or above on the KICD rubric for 'applying trigonometric ratios to angles of elevation and depression'.

		$15 \times \tan 27^\circ \approx 15 \times 0.5095 \approx 7.64$ m). Say: 'Both answers are close to 7.8 m. The small difference is rounding. What does this confirm about our method?' Wait 10 seconds. Cold-call: 'Wanjiku — what does it confirm?' Prompt learners to articulate: 'The height of the tank is the same regardless of where you stand — the ratio method is consistent.' Then ask: 'Where else around Kenya might engineers or surveyors use exactly this method?' Accept 3–4 responses (Thika Superhighway overpass heights, Mau Forest tree surveys, height of Kenyatta International Convention Centre, cliff heights in Hells Gate).		
Driving Question Board (DQB) Creation  VIDEO: Trigonometry review Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The teacher displays the class Driving Question Board: 'How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment without ever measuring it directly?' Each learner receives two sticky notes. On the first (green), they write one thing they can NOW explain about the water-tank problem that they could not explain in Lesson 1. On the second (orange), they write one new question or remaining puzzle. Learners post their notes and the teacher reads 4–5 aloud, grouping them into two columns: 'We can now explain...' and 'We still wonder...'. The teacher highlights that questions about depression angles, very steep angles (close to 90°), and using sine or cosine for height will be addressed in Lessons 4 and 5.	Uncover the DQB poster. Say: 'Cast your minds back to Lesson 1. Most of us said we had no idea how to find the tank height without climbing. Look at what you just did. You used a ratio — the tangent — to calculate it precisely from the ground. On your GREEN sticky note, write one thing you can NOW explain. On your ORANGE note, write one new question this lesson has sparked.' Give 3 minutes of silent writing. Then: 'Come and post them.' As learners post, read 4–5 green notes aloud, affirming each: 'Achieng writes: I can now explain why two students at different distances get the same answer — the ratio keeps the height constant. That is exactly right.' Read 2–3 orange notes aloud: 'Mwangi asks: what if the angle is very large, like 80°? Great question — the tangent of large angles gets very big, which means a steep line of sight to something very tall. We will explore that.' Physically move any sticky notes	Two-colour DQB reflection (Metacognitive progress monitoring). Requiring learners to explicitly articulate both growth ('I can now explain...') and remaining uncertainty ('I still wonder...') builds metacognitive awareness of their own learning trajectory. The public posting creates shared accountability and surfaces class-wide conceptual gaps for the teacher to address in upcoming lessons.	Read all orange sticky notes after class. Cluster them into categories. Three or more notes asking about the same confusion (e.g., 'when do I use sine instead of tangent?') flag a whole-class gap to address at the start of Lesson 4. Individual notes with idiosyncratic misconceptions (e.g., 'does the ratio change at night?') flag learners for one-on-one clarification. Green notes that are vague ('I understand trig now') suggest the learner may not yet have a precise conceptual grasp — follow up with a targeted question next lesson.

		from Lesson 1 that are now answered to a 'Resolved' section of the board.		
Model Building Phase  VIDEO: Finding reciprocal trig ratios Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners retrieve the annotated sketch of the water-tank scenario they drew in Lesson 1. They now upgrade it into a labelled mathematical model: they draw a clean right-angled triangle representing the scenario, label all three sides with their ROLE names (Opposite = height h, Adjacent = horizontal distance d, Hypotenuse = line of sight), write the three ratio equations ($\sin \theta = h/hyp$, $\cos \theta = d/hyp$, $\tan \theta = h/d$), and circle the equation that was sufficient to solve today's problem. Beneath the diagram they write a one-paragraph 'model statement' explaining, in their own words, how the ratio method works and why it is reliable. Learners who finish early add a second scenario: 'What if you knew the line-of-sight length instead of the horizontal distance — which ratio would you use then?'	Say: 'Turn to your Lesson 1 sketch of the water tank. We are going to upgrade it into a proper mathematical model. I want a clean triangle, all three sides labelled with their role names AND their ratio letters (O, A, H), and all three ratio equations written out. Then circle the one equation that solved our problem today.' Give 6 minutes. Circulate and narrate what you see: 'I can see Otieno has labelled the hypotenuse as the line of sight — that is precise and correct.' After 6 minutes, say: 'Now write your model statement underneath — one paragraph, in your own words: how does the ratio method work and why is it reliable? You have 4 minutes.' After writing, cold-call 2 learners to read their model statement aloud. Give specific praise for correct language: 'Notice how Halima used the word constant — the ratio is constant for a fixed angle. That is the key idea.' Close by pointing to the anchoring phenomenon poster: 'Next lesson we will discover something surprising about complementary angles — angles that add to 90° — and how their sine and cosine are secretly related. Keep your model — you will add to it.'	Cumulative annotated model revision. Requiring learners to physically upgrade the same diagram across every lesson creates a tangible record of conceptual growth. Labelling sides with both role names (Opposite, Adjacent, Hypotenuse) and letters (O, A, H) alongside ratio equations consolidates the vocabulary–symbol–meaning triangle that underpins all subsequent trigonometry work.	Collect the upgraded model diagrams at the end of class (or photograph them with a phone if printing is not possible). Assess against three criteria: (1) All three sides correctly labelled with role names relative to the marked angle; (2) All three ratio equations written correctly (not inverted); (3) Model statement uses the word 'ratio' and references the water-tank context specifically. Learners meeting all three criteria are at 'Meets Expectations'. Learners who add the extension scenario (sine for hypotenuse-known problems) are at 'Exceeds Expectations'.

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) During the measure-and-compute activity, did the majority of learners successfully identify opposite vs adjacent before measuring — or did confusion persist? If more than 30% of learners swapped the two sides, plan a 5-minute warm-up drill at the start of Lesson 4 using three quick labelling diagrams on the board. (2) When Student B's tank height (≈ 7.64 m) differed slightly from Student A's (≈ 7.81 m), did learners confidently attribute this to rounding, or did some express doubt in the method itself? If doubt was expressed, reinforce the idea that a real measurement with a precise clinometer would eliminate rounding error — connecting

forward to the Lesson 8 outdoor activity. (3) Were the orange DQB sticky notes asking substantive mathematical questions (about angle size, about which ratio to use, about depression angles) — or were they vague or off-topic? Substantive questions confirm learners are intellectually engaged with the phenomenon. Vague questions may indicate learners who need more concrete grounding before Lesson 4's complementary-angle work. (4) Did any group's measured ratios deviate significantly from the calculator values (e.g., off by more than 0.05)? If so, inspect their printed triangles — poor print quality or distortion can cause systematic measurement error; prepare a backup set of triangles drawn on squared paper for Lesson 4 if needed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners measured the sides of right-angled triangles with angles of 30°, 45°, and 60°, computed the three side ratios (O/H, A/H, O/A) for each, and verified their values matched the calculator's sin, cos, and tan outputs.
What did I learn?	The three trigonometric ratios — sine (opposite/hypotenuse), cosine (adjacent/hypotenuse), and tangent (opposite/adjacent) — are constant for a fixed angle regardless of the size of the triangle. The mnemonic SOH-CAH-TOA names and defines all three.
How does this explain the phenomenon?	Using the tangent ratio and the horizontal distance from the base of the school water-tank tower, learners calculated the tank height from two different positions and obtained consistent results — confirming that trigonometric ratios make precise indirect measurement possible without ever touching or climbing the structure.

LESSON 4 (40 minutes): The Constant Ratio — Discovering Sine, Cosine and Tangent Through Drawing

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners discover through hands-on drawing and measurement that the ratios opp/hyp, adj/hyp, and opp/adj remain constant for a fixed angle regardless of triangle size, and verify these ratios using a scientific calculator — directly explaining why any two observers at different distances from the water tank calculate the same height.
Knowledge	– The three trigonometric ratios — sine (opp/hyp), cosine (adj/hyp), and tangent (opp/adj) — are constant for a fixed acute angle, regardless of the size of the right-angled triangle. – A scientific calculator stores exact values of sine, cosine and tangent for every angle and can be used to verify hand-measured ratios. – The constancy of trigonometric ratios is the mathematical reason why two observers at different distances from the water tank arrive at the same height calculation.
Skills	– Draw accurate right-angled triangles with a specified acute angle using ruler and protractor, and measure sides to the nearest millimetre. – Compute sine, cosine and tangent ratios from measured side lengths, record results in a table, and compare against calculator values to evaluate accuracy.
Attitudes	– Develop confidence in using hands-on geometric construction as a valid route to mathematical discovery. – Appreciate that precision in measurement and careful reasoning lead to reliable, reproducible results — a value shared by Kenyan surveyors, engineers and scientists.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	Lessons 1–3 established the vocabulary and definitions of the three ratios (SOH-CAH-TOA). Lesson 4 is the pivotal discovery lesson: learners generate empirical evidence — by drawing and measuring three different-sized triangles all sharing the same 35° angle — that the ratios are truly constant. This 'aha' moment resolves the core puzzle of the phenomenon: it does not matter how far from the tank you stand, the ratio $\tan(\theta) = \text{height} \div \text{distance}$ is always the same, so the height is always recoverable.
Safety Notes	Compass points and sharp pencils are used during construction. Remind learners to keep compass tips pointed downward when not in use and to pass rulers with the flat edge forward. No other specific hazards.

B. LESSON OVERVIEW

This lesson is the empirical heart of the trigonometry sequence. Having spent Lessons 1–3 naming the sides of a right-angled triangle and defining the three ratios conceptually, learners now gather their own physical evidence. Working in pairs, each pair uses a ruler and protractor to construct three right-angled triangles that all contain a 35° angle but differ in the length of their adjacent side (5 cm, 8 cm, and 11 cm). They label the opposite, adjacent and hypotenuse for each triangle, measure all three sides carefully, and enter the values into a structured data table. They then calculate opp/hyp, adj/hyp and opp/adj for each triangle. The repeated observation that these three columns remain essentially constant across all three triangles — despite the triangles being different sizes — is the central discovery of the lesson.

In the Explain Phase, the teacher names the three constant ratios as sine, cosine and tangent, anchors the mnemonic SOH-CAH-TOA from Lesson 3, and guides learners to verify their hand-measured values against the calculator's $\sin(35^\circ)$, $\cos(35^\circ)$ and $\tan(35^\circ)$. Any small discrepancies are attributed to measurement error rather than mathematical inconsistency, reinforcing the idea that the calculator holds the 'true' ratio. The connection to the water-tank phenomenon is made explicit: because

$\tan(35^\circ)$ is always the same number, a student standing 6 m from the tank and a student standing 10 m from the tank will both calculate the same height, provided they measure their angle correctly — which is exactly why the method works.

The lesson closes with learners revising their cumulative model diagram to annotate which ratio links the ground distance (adjacent), the tank height (opposite) and the line of sight (hypotenuse) to the observed angle. The Driving Question Board receives two new sticky notes: one recording the discovered constant and one posing the next question — what if the angle changes? This naturally sets up Lesson 5, which explores complementary angles and the sine–cosine relationship.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Intro to the trigonometric ratios Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner opens their exercise book to a fresh page and individually writes a prediction in response to the prompt displayed on the board: 'If I draw three right-angled triangles — all with a 35° angle but different sizes — and I calculate opp÷hyp for each one, will I get the same answer, a bigger answer each time, or a smaller answer each time? Write your prediction AND your reason in one sentence.' Learners write silently for 2 minutes, then share briefly with their desk partner before the teacher takes two or three cold-call responses.	Write the prompt on the board before learners enter: 'Three triangles. Same angle (35°). Different sizes. Same ratio or different ratios?' As learners settle, say: 'Before we touch a ruler today, I want your brain's best guess. Write your prediction and your reason — one sentence each — in your exercise book. You have two minutes. Work alone.' After 2 minutes, say: 'Turn and tell your partner your prediction in exactly one sentence. Go.' Allow 45 seconds. Then cold-call 3 learners (aim for one 'same', one 'different', one 'unsure'): 'Akinyi, what did you predict?' — 'Otieno, did you agree with Akinyi or disagree?' — 'Wanjiru, what was your reason?' Do NOT confirm or deny predictions. Say: 'We are going to let the triangles themselves answer that question. Keep your prediction visible — we will return to it.' [WAIT TIME: 10 seconds after each cold-call before prompting further.]	Predict–Observe–Explain (POE) activation: by committing a prediction to paper before the activity, learners create a cognitive stake in the outcome. When the observation contradicts an incorrect prediction, the conflict drives deeper sensemaking. Recording the reason (not just the guess) surfaces prior conceptions the teacher can monitor during the Observe Phase.	Teacher scans written predictions as learners share with partners — look for the proportion of learners predicting 'different ratios' (the common misconception that bigger triangles produce bigger ratios). This informs pacing: if more than half predict 'different', spend an extra 30 seconds at the start of the Observe Phase reminding learners that in Lesson 3 they defined the ratio relative to the angle, not the size.
Observe Phase  VIDEO: Finding reciprocal trig ratios Source: Khan	Working in pairs, learners receive a printed or teacher-drawn construction guide sheet and follow a four-step protocol. Step 1 — each pair draws three right-angled triangles on plain A4 paper:	Distribute pre-ruled data tables and say: 'You are going to be mathematicians collecting your own evidence right now — just like surveyors in the Rift Valley who measure land before building	Data-table construction with pattern identification: the structured table forces learners to compute the same ratio three times for the same angle, making the pattern of constancy visible	Teacher checks three specific indicators while circulating: (1) Is the 35° angle correctly placed at the base vertex, not at the top? (2) Are learners correctly identifying the hypotenuse as the longest

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Triangle A with adjacent side = 5 cm, Triangle B with adjacent side = 8 cm, and Triangle C with adjacent side = 11 cm. In each triangle the 35° angle is at the left-hand base vertex and the right angle is at the right-hand base vertex. Step 2 — using a ruler, learners measure the opposite side (vertical) and the hypotenuse (slanted) for each triangle to the nearest millimetre. Step 3 — learners complete a class data table (columns: Triangle, Opposite, Adjacent, Hypotenuse, opp÷hyp, adj÷hyp, opp÷adj) by dividing the measured values on a calculator and rounding to 2 decimal places. Step 4 — learners look down each ratio column and write one observation sentence at the bottom of the table.	anything.' Demonstrate Triangle A construction on the board in 90 seconds: draw a horizontal base, mark the left vertex, use a protractor to mark exactly 35° above the base, draw the adjacent side (5 cm) along the base, erect a vertical perpendicular at the right end, then connect to the 35° ray to complete the triangle. Say: 'The angle goes here — 35° — and your right angle is here. Measure opposite and hypotenuse with your ruler. Every millimetre counts.' Circulate during construction (approximately 8 minutes). Targeted questions while circulating: 'Which side is your hypotenuse — how do you know?' and 'Your adjacent is exactly 8 cm — so where does your ruler start?' After pairs complete the table, pause the class: 'Look at your opp÷hyp column. What do you notice? Write one sentence.' [WAIT TIME: 15 seconds of silent writing.] Cold-call 4 pairs: 'Kamau and Nafula — what did you write?' — 'Omondi and Chebet — does your table agree or disagree with them?' Anticipated response: all values in each column are approximately equal. If a pair's values diverge, ask: 'Could measurement error in your ruler explain the difference?' Introduce the idea that small discrepancies (~0.02) are measurement error, not mathematical error.	without the teacher stating it. The 'one observation sentence' prompt pushes learners to articulate the pattern in their own words before the teacher provides the formal name — this is the Notice and Wonder routine embedded into a data-collection task.	side, not the vertical side? (3) Do the computed ratios in each column vary by no more than ±0.03 (acceptable measurement error)? Any pair with ratios diverging by more than ±0.05 likely has a protractor-reading error — teacher asks them to re-measure the angle. A quick whole-class check: 'Raise your hand if your three opp÷hyp values are all between 0.55 and 0.60.' If fewer than 70% raise hands, pause and re-examine one triangle together on the board.
Explain Phase VIDEO: The trig functions & right triangle trig ratios Source: Khan Academy (English - US curriculum) Search ARES	Learners use their scientific calculators to find sin(35°), cos(35°) and tan(35°) and compare these to their table values. They record the calculator values in a new row labelled 'Calculator (exact)' at the bottom of the table. They then write a	Say: 'Your table gives you an experimental value of opp÷hyp. Now let us see what mathematics says the exact value should be. On your calculator, press: sin, 35, equals. Write that number.' [WAIT TIME: 10 seconds.] Cold-call: 'Nyambura, what did you get?'	Claim–Evidence–Reasoning (CER) writing: the conclusion statement ('sin = ___, cos = ___, tan = ___, and these values do not change because ___') requires learners to move from observed data (evidence) to a causal explanation (reasoning). The two-	Collect the conclusion statements from 6–8 learners (select a stratified sample: 2 strong, 4 mid, 2 struggling). Look for: (1) correct decimal values matching the calculator; (2) a 'because' clause that references the angle being fixed, not the triangle size; (3)

for similar videos EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	conclusion statement: 'For a 35° angle, $\sin = \underline{\hspace{1cm}}$, $\cos = \underline{\hspace{1cm}}$, $\tan = \underline{\hspace{1cm}}$, and these values do not change because $\underline{\hspace{1cm}}$.' Learners are then directed back to the water-tank scenario: using $\tan(35^\circ)$ from the calculator and a horizontal distance of 6 m, they calculate the height of the tank. A second calculation uses the same $\tan(35^\circ)$ but a distance of 10 m. Learners verify that both calculations produce the same height (to confirm consistency) before answering: 'Why do two observers at different distances still get the same answer?'	Expected: 0.5736. 'How close is that to your measured value?' Then: 'Press cos, 35, equals.' Cold-call: 'Kipchoge, what is cosine 35?' Expected: 0.8192. 'And tan, 35, equals?' Expected: 0.7002. Write on the board: $\sin(35^\circ) = 0.5736$, $\cos(35^\circ) = 0.8192$, $\tan(35^\circ) = 0.7002$. Say: 'These are the exact ratios. Your ruler gave you something very close. That is not luck — that is geometry being consistent.' Now say: 'Back to our water tank. Imagine Zawadi stands 6 metres from the base of the tank and measures an angle of elevation of 35°. Write: $\tan(35^\circ) = \text{height} \div 6$. Solve for height.' [WAIT TIME: 15 seconds.] 'Now Baraka stands 10 metres away and also measures 35°. Write: $\tan(35^\circ) = \text{height} \div 10$. Solve.' Cold-call two learners to give answers. Say: 'Different distances, same angle — do you get the same height?' [Expected: Yes, both give the same height because $\tan(35^\circ)$ is constant.] 'This is EXACTLY why the water-tank method works. The ratio locks the height to the distance. Change the distance, the height changes proportionally — but the ratio stays constant.' Reinforce SOH-CAH-TOA by pointing to the board and having the class recite it once chorally.	observer water-tank calculation is a transfer task that applies the new understanding to the anchoring phenomenon, closing the explanatory loop opened in Lesson 1.	correct tan calculation for the water-tank problem (height = $\tan(35^\circ) \times \text{distance}$). Red flag: learners who write 'the ratio stays the same' — these learners have not yet grasped that size can change while angle is held constant. Flag for targeted follow-up at the start of Lesson 5.
Driving Question Board (DQB) Creation VIDEO: Trigonometry review Source: Khan Academy (English - US curriculum)	Each learner writes two sticky notes (or paper slips). The first is a 'We now know' note: one sentence stating the key discovery from today's lesson. The second is a 'We still wonder' note: a question that today's lesson raised but did not answer. Pairs share their notes briefly, then one representative from each group of four posts the	Distribute two paper slips (or direct learners to tear a page). Say: 'Slip 1 — green side: write one thing you are now certain about that you were not certain about at the start of this lesson. Start with: We now know...' [WAIT TIME: 60 seconds of silent writing.] 'Slip 2 — yellow side: write the most interesting question today's lesson raised for	Two-column Driving Question Board (Known / Wondering): the DQB makes the cumulative storyline visible to learners as a physical record on the classroom wall. Posting 'We now know' statements consolidates today's learning; posting 'We still wonder' questions maintains productive intellectual momentum and gives	Teacher reads all posted 'We now know' slips after class. Success indicator: slips reference constancy of ratios and/or the angle as the determining factor. Concern indicator: slips that say only '$\sin = 0.57$' without any causal claim — these learners have memorised a value but not grasped the constancy principle.

Search ARES for similar videos EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	notes on the class DQB under the columns 'What we have figured out' and 'What we still need to figure out'. The teacher reads two or three 'wonder' questions aloud and explicitly connects one to the next lesson.	you. Start with: 'We still wonder...' [WAIT TIME: 60 seconds.] Ask one member from each pair to post their slips on the DQB. Read three 'wonder' slips aloud. Likely student questions: 'What happens to sin and cos if the angle changes to 40°?', 'Are sin and cos of 35° and 55° related?', 'How do we use this if we do not know the angle — only the sides?' Say: 'These are exactly the questions we will chase in Lessons 5 and 6. Notice this one — [point to a sine/cosine of complementary angles question if present] — we will investigate that next lesson with a discovery that surprises most mathematicians the first time they see it.' Update the DQB headline with today's addition: 'The three trig ratios are CONSTANT for a fixed angle — so any distance works.'	learners ownership of the inquiry direction. The teacher's act of reading wonder questions aloud validates student curiosity as a legitimate part of mathematics.	The 'wonder' slips also serve as a needs-assessment for Lesson 5: if many students ask about what happens when the angle changes, the complementary-angle investigation in Lesson 5 will land with high engagement.
Model Building Phase VIDEO: Intro to the trigonometric ratios Source: Khan Academy (Kiswahili) Search ARES for similar videos EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners return to the cumulative annotated diagram they began in Lesson 1 (the water-tank right-angled triangle with the observer on the ground, the vertical tank height, and the line of sight). Today they add three specific annotations: (1) they label the three sides as H (hypotenuse = line of sight), O (opposite = tank height) and A (adjacent = ground distance) as before, but now (2) they write the three ratio equations inside or beside the triangle: $\sin \theta = O/H$, $\cos \theta = A/H$, $\tan \theta = O/A$. (3) They write a starred note in a box: '★ These ratios are CONSTANT for a fixed θ — proven by our 3-triangle experiment today.' Finally, learners update the bottom of their model page with a one-sentence 'model status': 'I can now calculate the water-tank height if I know the	Say: 'Open your model diagram from Lesson 1. We have been building this picture of the water-tank problem since day one. Today we made a major discovery that upgrades the model.' Display your own annotated version on the board (or a poster). Walk through the three annotations: 'First, confirm your side labels — O, A, H — relative to angle θ at the observer's position. Second, write the three equations: $\sin \theta = O \text{ over } H$, $\cos \theta = A \text{ over } H$, $\tan \theta = O \text{ over } A$. Third — this is the most important upgrade — write this starred note in a box: these ratios are constant for a fixed θ.' Model writing the box on the board. Then say: 'Finally, write a model status sentence at the bottom: what can you NOW do that you could NOT do at the start of Lesson 1?' [WAIT TIME: 90 seconds of silent writing.]	Cumulative Model Revision: the model diagram functions as a running scientific model that learners update each lesson as their understanding grows. This technique — borrowed from NGSS Science and Engineering Practice 2 (Developing and Using Models) — makes conceptual growth visible and gives learners a concrete artefact that spans the entire 8-lesson sequence. The 'model status' sentence forces metacognitive reflection on what they can now do, which consolidates procedural fluency alongside conceptual understanding.	Teacher circulates during model annotation and checks: (1) Are all three ratio equations correctly written with O, A, H labelled consistently? (2) Is the starred constancy note present? (3) Does the model status sentence include both the formula ($\tan \theta \times \text{distance}$) and a reference to the water tank? Exit-ticket equivalent: before learners close their books, teacher asks the whole class to point to which ratio they would use to find the tank height if they know the ground distance and the angle. If more than 20% point to sine or cosine rather than tangent, spend the first 3 minutes of Lesson 5 doing a rapid recap of which sides are involved in each ratio relative to the water-tank triangle.

	ground distance and the angle of elevation, using $\text{height} = \tan(\theta) \times \text{distance}$.'	Cold-call 3 learners to read their model status sentences. Look for: ability to state the calculation as $\text{height} = \tan(\theta) \times \text{distance}$. Close by saying: 'Your model is growing. In Lesson 1 it was just a picture. Now it has equations, constant ratios, and a calculation strategy. Two more lessons and your model will be complete enough to go outside and measure a real structure.'		
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D. TEACHER REFLECTION

1. When learners computed the three ratio columns in their data tables, did the values cluster tightly enough (within ± 0.03) to make the constancy pattern convincing, or did measurement errors obscure the discovery? What construction or measurement scaffolds could reduce error in future?
2. Which specific misconception was most prevalent during cold-calls in the Predict Phase — that bigger triangles produce bigger ratios, or that the ratio depends on which side you measure first? How will you address this in the opening of Lesson 5?
3. The two-observer water-tank calculation ($\tan 35^\circ$ at 6 m vs. 10 m) was intended as a transfer task. Did learners make the connection spontaneously or did it require significant teacher prompting? What does this tell you about the depth of their understanding of ratio constancy?
4. Review the 'We still wonder' DQB slips: were learners' questions productively forward-looking (e.g., about complementary angles or changing angles), or were they mostly clarification questions about today's content? What does the balance tell you about lesson pacing?
5. During the Model Building Phase, how precise and mathematically complete were learners' model status sentences? Did they use the formula $\text{height} = \tan(\theta) \times \text{distance}$, or did they write only a vague description? Plan a targeted starter activity for Lesson 5 if fewer than two-thirds of the class could articulate the formula independently.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We drew three right-angled triangles — all with a 35° angle but with adjacent sides of 5 cm, 8 cm and 11 cm — and measured the opposite side and hypotenuse in each. We calculated $\text{opp} \div \text{hyp}$, $\text{adj} \div \text{hyp}$ and $\text{opp} \div \text{adj}$ for all three triangles and recorded the values in a table.
What did I learn?	The three ratios (opp/hyp , adj/hyp , opp/adj) stayed almost exactly the same across all three triangles, even though the triangles were different sizes. The calculator confirmed the exact values: $\sin(35^\circ) \approx 0.574$, $\cos(35^\circ) \approx 0.819$, $\tan(35^\circ) \approx 0.700$. These constant ratios are called sine, cosine and tangent (SOH-CAH-TOA).
How does this explain the phenomenon?	Because $\tan(\theta)$ is constant for a fixed angle, any observer — whether standing 6 m or 10 m from the base of the water tank — will calculate the same height using $\text{height} = \tan(\theta) \times \text{ground distance}$. The constancy of the ratio is the mathematical reason the water-tank measurement method works from any distance.

LESSON 5 (80 minutes): Sine and Cosine of Complementary Angles

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to discover and apply the complementary-angle relationship between sine and cosine, and use it to deepen their understanding of how a single right-angled triangle can reveal multiple unknown measurements — including the height of the school water tank.
Knowledge	Learners identify that for any two complementary angles (summing to 90°) in a right-angled triangle, the sine of one angle equals the cosine of the other ($\sin \theta = \cos(90^\circ - \theta)$), and explain why this symmetry arises from the fixed ratios of sides in the triangle.
Skills	Learners generate a table of complementary-angle pairs, compute their sines and cosines using a calculator or mathematical tables, identify the pattern, and apply the shortcut to solve missing-angle and missing-side problems involving the water tank scenario.
Attitudes	Learners appreciate the elegant internal consistency of trigonometric ratios — that two students standing at different distances from the water tank, each measuring a different angle, will always arrive at the same height — demonstrating that mathematics is reliable, not accidental.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	In Lessons 3 and 4, learners defined sin, cos, and tan and used them to calculate the water tank's height from a known distance and angle. In this lesson (Lesson 5), learners discover that $\sin \theta = \cos(90^\circ - \theta)$, unlocking a powerful shortcut: if you know the angle of elevation from one student's position, you can instantly find the complementary angle's trig ratio without re-measuring — and the height answer remains consistent. This prepares learners for Lesson 6 (angles of depression) and the outdoor clinometer activity in Lesson 8.
Safety Notes	No outdoor activity in this lesson. Calculators and mathematical tables are handled with care. Learners are reminded not to share personal digital devices without permission and to observe online safety when using any digital resource.

B. LESSON OVERVIEW

Learners begin by revisiting the water tank scenario: two students stood at different distances and measured different angles of elevation yet calculated the same height. The teacher challenges learners to notice what happens when the two angles are complementary (add to 90°). Through a structured table-building activity, learners compute sin and cos values for complementary pairs (e.g., 30° & 60° , 25° & 65° , 40° & 50°) and discover that $\sin \theta = \cos(90^\circ - \theta)$. They formalise this relationship, practise applying it to solve problems without a calculator, and update their Driving Question Board and cumulative triangle model to record this new explanatory tool.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Trig word problem: complementary	Learners are shown a diagram of the school water tank on the board. Two students — Akinyi standing 8 m from the base (angle of elevation = 60°) and Baraka	Display the water tank diagram on the board (or draw it clearly with labelled angles 30° and 60°). Say: 'In our last two lessons we used tan, sin, and cos to find the tank's	Predict-first protocol (Science and Engineering Practice: Asking Questions). By committing a prediction to paper before seeing data, learners activate prior	Teacher circulates and reads exercise books during the 3-minute writing window. Observable indicator: learner writes a complete sentence that references at least

angles Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Sine and Cosine of Complementary Angles (Flexbook) Source: CK-12 Search ARES for similar readings	standing further away (angle of elevation = 30°) — are marked. The teacher asks: 'Akinyi's angle and Baraka's angle add up to 90°. Before we calculate anything — do you think their sine and cosine values are connected in some special way? Write your prediction in your exercise book using the sentence stem: I predict that $\sin 60^\circ$ and $\cos 30^\circ$ are _____ because _____. Learners write individually for 3 minutes, then share with a shoulder partner for 1 minute.	height. Today we zoom in on something surprising about complementary angles — angles that add to 90°.' Distribute the sentence-stem prompt card or write it on the board. After 3 minutes of silent writing, cold-call 3 learners to share predictions. Record each prediction verbatim on the board under the heading 'Our Predictions'. Do NOT confirm or deny. Say: 'We will return to these predictions at the end of the lesson to see who was right — and why.' WAIT TIME: allow 12 seconds of silence after posing the question before calling on anyone.	knowledge of sin and cos ratios and create cognitive dissonance that the observation phase will resolve. Recording predictions publicly creates accountability and motivation.	one trig ratio (sin or cos) and gives a reason — even if the reason is incorrect. Learners who write only a number without reasoning are prompted: 'Can you say why you think that?'
Observe Phase  VIDEO: Sine & cosine of complementary angles Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Sine and Cosine of Complementary Angles Practice Source: CK-12 Search ARES for similar readings	Learners work in groups of 3–4 to complete a structured 'Complementary Angles Table'. Each group receives (or draws) a table with columns: Angle θ $90^\circ - \theta$ $\sin \theta$ $\cos \theta$ $\sin(90^\circ - \theta)$ $\cos(90^\circ - \theta)$. They fill in angle pairs: (20°, 70°), (30°, 60°), (35°, 55°), (40°, 50°), (45°, 45°), (50°, 40°), (60°, 30°). Using mathematical tables or calculators, groups compute all six columns for each pair, record values to 4 decimal places, and then annotate: 'What do you notice about $\sin \theta$ and $\cos(90^\circ - \theta)$? What about $\cos \theta$ and $\sin(90^\circ - \theta)$?' Groups write one group observation statement at the bottom of their table.	Distribute the blank table template (or instruct learners to draw it — allow 2 minutes). Say: 'Use your mathematical tables or calculator. Fill in every cell carefully. When you are done, look across each row and across the columns. Describe in words what pattern you see — be as specific as possible.' Circulate to each group. Prompt groups that are stuck: 'Look at $\sin 30^\circ$ and $\cos 60^\circ$. Are those numbers familiar to each other?' After 15 minutes of group work, pause the class. Say: 'Without giving the answer away — raise your hand if your group found a surprising match somewhere in the table.' Cold-call 4 groups to share ONE observation each. Write key observations on the board. WAIT TIME: 10 seconds after asking each group for their observation before moving to the next group.	Data-pattern recognition through a structured table (Science and Engineering Practice: Analysing and Interpreting Data). The table is designed so the pattern $\sin \theta = \cos(90^\circ - \theta)$ becomes visually obvious across every row. The act of computing values themselves — rather than being told the formula — means learners own the discovery. The (45°, 45°) row is a deliberate anchor: $\sin 45^\circ = \cos 45^\circ$, which is a special, memorable case.	Teacher checks group tables for accuracy in at least 3 rows before the whole-class share. Observable indicator: all values in the $\sin \theta$ and $\cos(90^\circ - \theta)$ columns match for every row (to 2 decimal places minimum). Groups with mismatches are directed to recheck their calculator input — teacher models pressing the correct key sequence once if needed.
Explain Phase  VIDEO: Finding reciprocal trig ratios	The class co-constructs the rule. The teacher draws a single right-angled triangle on the board — the same water tank triangle from	Draw the water tank triangle slowly and deliberately. Say: 'Let us go back to our tank. The angle at Akinyi's feet is θ. What is the angle	Label-swap reasoning (Science and Engineering Practice: Constructing Explanations). By physically re-labelling opposite and	Observe learner responses to Problem 1 (purely symbolic — no triangle). Observable indicator: learner correctly writes $\cos 53^\circ =$

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Lesson 3 — with angle θ at the base (Akinyi's position) and the complementary angle $(90^\circ - \theta)$ at the top. Learners are guided to label: opposite and adjacent sides relative to θ, then re-label relative to $(90^\circ - \theta)$. They write: $\sin \theta = \text{opposite/hypotenuse}$ and $\cos(90^\circ - \theta) = \text{adjacent of } (90^\circ - \theta)/\text{hypotenuse}$ — and discover these are the same fraction. The teacher then poses three application problems using the water tank: (1) If $\sin 37^\circ = 0.6018$, write down $\cos 53^\circ$ without using a calculator. (2) If the angle of elevation from Akinyi's position is 62°, what is the complementary angle, and what does $\sin 62^\circ$ equal in terms of cosine? (3) Baraka measures an angle of elevation of 28°. Akinyi's angle is the complement. If the tank is 6 m tall, use $\sin 28^\circ = 0.4695$ and the complementary rule to find $\cos 62^\circ$ and verify. Learners solve individually, then compare with a partner.	at the top of the triangle — at the tip near the tank top? Give me 12 seconds to think.' [WAIT 12 seconds.] Cold-call 2 learners. Confirm: '90° minus θ — because all angles in the triangle sum to 180° and one is already 90°.' Guide label-swapping: 'If I stand at the top angle instead of Akinyi's angle, which side is now the opposite? Which is the adjacent?' After the discovery, write the rule formally: $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$. Say: 'This is not magic — it is because the same two sides are involved, just from a different angle's point of view.' Post the three application problems. After 8 minutes of individual work, cold-call 3 different learners (one per problem) to present at the board. Correct misconceptions immediately with: 'The sides did not change — only which angle we are looking from changed.'	adjacent relative to the complementary angle on the same diagram, learners see that the rule is not a coincidence but a direct consequence of the triangle's fixed geometry. Connecting this back to the water tank makes the abstraction concrete and meaningful.	0.6018 and gives the reason 'because $37^\circ + 53^\circ = 90^\circ$'. Learners who give a different decimal are asked to re-examine their table from Phase 2. Problem 3 serves as a cross-check: if learners get $\cos 62^\circ \neq 0.4695$, teacher diagnoses whether the error is in the complementary rule or in arithmetic.
Driving Question Board (DQB) Creation VIDEO: Using complementary angles Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Trig values of special angles Source: Khan Academy (English - US curriculum)	Each learner writes one sticky note (or index card) with: (a) one thing they can NOW explain about the water tank that they could not explain in Lesson 1, and (b) one new question the lesson has raised. Sticky notes are placed on the class Driving Question Board under two columns: 'We can now explain...' and 'We still wonder...'. The class does a 3-minute silent gallery walk to read peers' cards. The teacher then reads aloud two 'We can now explain' entries and two 'We still wonder' entries, and briefly maps them onto the lesson sequence: 'Lessons 6 and 7 will answer the depression questions; Lesson 8 will let you test	Distribute sticky notes or index cards. Say: 'You have 4 minutes. Write neatly — your classmates will read these.' After posting, say: 'Walk around quietly for 3 minutes. Read at least five cards that are not yours. Place a small tick on any card where you had the same thought.' After the gallery walk, stand at the DQB and read two entries from each column aloud. Say: 'Notice how many of you are now asking about what happens when you look DOWN from the tank — that is called the angle of depression, and it is Lesson 6. Keep that curiosity alive.' Map the DQB visually: draw a simple timeline of the 8-lesson sequence	Public knowledge tracking (Science and Engineering Practice: Obtaining, Evaluating, and Communicating Information). The DQB makes the class's growing understanding visible and cumulative. The gallery walk builds metacognitive awareness — learners see that peers share their confusions, normalising productive struggle. Mapping remaining questions to future lessons sustains the inquiry arc.	Teacher reads all sticky notes after the lesson. Observable indicator: at least 80% of learners' 'We can now explain' entries correctly reference the complementary-angle relationship (not just 'I understand sine better'). Vague entries are flagged for follow-up at the start of Lesson 6.

Search ARES for similar readings	everything outside at the real tank.'	on the board and mark 'We are HERE — Lesson 5' with an arrow.		
Model Building Phase VIDEO: Intersection points of $y=\sin(x)$ and $y=\cos(x)$ Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Relate ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners retrieve their cumulative right-angled triangle model (a diagram started in Lesson 3 and updated in Lesson 4). They add two new annotations in a distinct colour: (1) Label the two acute angles as θ and $(90^\circ - \theta)$ and draw a curved double-headed arrow between them labelled 'complementary pair'. (2) Write beside the hypotenuse: $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$ with a short note explaining what this means in the water tank context — e.g., 'Akinyi's $\sin$ = Baraka's $\cos$ when their angles add to 90° .' Learners then solve one final consolidation problem in their model diagram: 'The water tank is 7.5 m tall. Student Chemutai stands so her angle of elevation is 48° . Student Otieno's angle of elevation is the complement. Without using a calculator, write down $\sin 48^\circ$ in terms of a cosine. If $\sin 48^\circ = 0.7431$, find Otieno's horizontal distance from the tank base.' (Answer: $\cos 42^\circ = 0.7431$; distance = $7.5 \div \tan 42^\circ$ — teacher confirms this bridges to next lesson.)	Say: 'Take out your triangle model. We are going to make it even more powerful today.' Give learners 5 minutes to add the complementary-angle annotations. Circulate and check that arrows and equations are correctly placed. After 5 minutes, pose the Chemutai-Otieno problem. Say: 'You have 6 minutes — work individually. The first step is to identify the complementary pair. Do that first, then use our new rule.' After 6 minutes, cold-call 2 learners to present their working on the board step by step. Affirm correct steps and correct errors. Close with: 'Every time we add to this model, the water tank puzzle becomes easier to solve. In Lesson 6 we add one more tool — the angle of depression — and in Lesson 8 you will take this model outside and solve the real thing.' WAIT TIME: 12 seconds after posing the consolidation problem before allowing any questions.	Cumulative model revision (Science and Engineering Practice: Developing and Using Models). Each lesson's core insight is inscribed into the evolving triangle diagram, creating a physical, visual record of the explanatory journey. By solving a new problem using the model, learners practise transferring the complementary-angle rule to a novel but familiar context — the water tank — reinforcing both procedural fluency and conceptual understanding.	Collect or photograph learner models at the end of the lesson. Observable indicators: (1) Complementary-angle label and double-headed arrow correctly placed on both acute angles. (2) $\sin \theta = \cos(90^\circ - \theta)$ written correctly beside or below the triangle. (3) Consolidation problem: learner correctly identifies $\cos 42^\circ = 0.7431$ and sets up $7.5 \div \tan 42^\circ$ for Otieno's distance (even if final arithmetic is incomplete, the setup earns credit). Learners missing all three indicators receive a targeted re-explanation card at the start of Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Did at least 75% of learners correctly complete the complementary-angles table with matching $\sin \theta$ and $\cos(90^\circ - \theta)$ values in Phase 2? If not, consider starting Lesson 6 with a 5-minute re-do of two rows using a think-aloud. (2) Were learners able to articulate WHY $\sin \theta = \cos(90^\circ - \theta)$ (label-swap reasoning) or did they treat it as a rule to memorise? If the latter, revisit the triangle diagram at the start of Lesson 6 and re-ask: 'Which side is opposite depends on WHICH angle you are standing at — the sides did not move.' (3) Did the DQB reveal any persistent misconceptions (e.g., learners confusing complementary with supplementary angles)? Plan a 2-minute clarification at the start of Lesson 6. (4) Did the Chemutai-Otieno consolidation problem stretch learners appropriately — was it too easy (most finished in under 3 minutes) or too hard (fewer than half set it up correctly)? Adjust the difficulty of the opening warm-up in Lesson 6 accordingly. (5) Note which learners were cold-called and ensure different learners are called in Lesson 6 to maintain equitable participation across gender and seating position.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners computed sine and cosine values for complementary angle pairs (e.g., 30° & 60°, 40° & 50°, 45° & 45°) using mathematical tables or calculators, recording results in a structured table, and identified that $\sin \theta = \cos(90^\circ - \theta)$ across every row.
What did I learn?	For any two complementary angles in a right-angled triangle, the sine of one angle equals the cosine of the other — because the same pair of sides is involved, just viewed from different angle positions. This means $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$.
How does this explain the phenomenon?	In the water tank scenario, if Akinyi's angle of elevation (θ) and Baraka's angle of elevation ($90^\circ - \theta$) are complementary, then $\sin(\text{Akinyi's angle}) = \cos(\text{Baraka's angle})$ — confirming that the two students' measurements are internally consistent and that the same height can be recovered from either position using this shortcut.

LESSON 6 (80 minutes): Sines and Cosines of Complementary Angles — The Hidden Shortcut in the Tank Problem

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners discover and explain the complementary-angle relationship $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$, connecting it directly to the geometry of the water-tank right-angled triangle.
Knowledge	The learner identifies that in a right-angled triangle the 'opposite' side for one acute angle is the 'adjacent' side for the other acute angle, and deduces that $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$ for all acute angles θ .
Skills	The learner generates and analyses a systematic table of complementary angle pairs, constructs a labelled right-angled triangle diagram to justify the pattern, and applies the relationship to simplify trigonometric expressions and solve problems involving angles of elevation.
Attitudes	The learner appreciates the elegance and consistency of mathematics — that two students at different distances from the water tank, using supplementary angles, obtain the same height — and develops confidence in using logical reasoning to reveal hidden structure.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	In Lessons 1–5 learners built the sin, cos, and tan ratios and used them to calculate the height of the school water tank from a single angle of elevation. In this lesson they discover that the two acute angles inside their right-angled tank diagram are complementary (they sum to 90°), and that swapping sin and cos across those angles always gives the same result. This 'shortcut' deepens their understanding of why two students standing at different distances get different elevation angles yet calculate the SAME tank height — a key unresolved puzzle on the Driving Question Board — and equips them for the clinometer fieldwork in Lessons 7 and 8.
Safety Notes	No hazardous materials are used. Remind learners to handle geometrical set compasses with care. If learners move desks to work in groups, ensure aisles remain clear.

B. LESSON OVERVIEW

Learners begin by re-examining the water-tank right-angled triangle diagram built in earlier lessons and making a prediction about what happens to the sine and cosine values when they switch between the two acute angles. Working in pairs, they complete a structured table of complementary angle pairs (e.g., 20° & 70° , 35° & 55° , 45° & 45°) using pre-calculated or calculator-derived values, identify the pattern $\sin \theta = \cos(90^\circ - \theta)$, and then construct an annotated right-angled triangle to explain WHY the pattern holds — tracing how the 'opposite' and 'adjacent' labels flip when the reference angle changes. The lesson closes with learners updating the Driving Question Board and building a revised cumulative model of the tank scenario that now incorporates the complementary-angle insight.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Trig word problem: complementary	Learners look at the class anchor diagram (the labelled right-angled triangle showing a student on the ground, the horizontal distance to the base of the water-tank tower,	Display the anchor diagram on the board (or sketch it clearly). Point to the angle of elevation at the student's feet and then trace your finger to the angle at the top of the	Think-Pair-Share with public prediction display. By externalising predictions before instruction, learners activate prior knowledge of angle sums in a triangle (from	Observable indicator: Can learners state that angles in a triangle sum to 180° , and therefore the two acute angles in a right-angled triangle sum to 90° ? Listen for this

angles Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	the vertical height, and the angle of elevation θ). The teacher reveals that there is a second acute angle at the top of the triangle — the angle between the height and the hypotenuse — and asks learners to record individual silent predictions to two questions: (1) 'What do the two acute angles in our tank triangle add up to?' and (2) 'If I know $\sin 35^\circ$, can I immediately say anything about $\cos 55^\circ$ — without a calculator?' Learners write their predictions in their exercise books and share with a partner (Think-Pair) for 2 minutes before the teacher takes 4 cold-calls.	triangle. Say: 'We have been focused on this angle down here — the one our clinometer will measure when we stand outside next week. But there is another angle hiding up at the top of our triangle. Take 60 seconds of SILENT thinking time — do not write yet — just look.' [WAIT: 60 seconds, no talking.] Then say: 'Now write your predictions to both questions in your book. You have 90 seconds.' [WAIT: 90 seconds.] Then: 'Share with your neighbour. You have 2 minutes.' [WAIT: 2 minutes.] Cold-call exactly 4 learners — select one who looks uncertain. Record ALL predictions on the board without evaluating them. Say: 'We will return to these at the end to see who was closest.'	earlier grades) and create a personal stake in the outcome, priming them to notice the pattern in the Observe phase.	language during cold-calls. Learners who say 'the angles add to 90°' are ready for the table activity; those who say '180°' or are unsure need the teacher to briefly re-establish the right-angle constraint before proceeding.
Observe Phase VIDEO: Sine & cosine of complementary angles Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Sine and Cosine of Complementary Angles (Flexbook) Source: CK-12 Search ARES for similar readings	Working in pairs, learners complete a structured two-page investigation activity titled 'The Complementary Angle Table.' Part A: They fill in a table of 7 complementary angle pairs — $(10^\circ, 80^\circ)$, $(20^\circ, 70^\circ)$, $(30^\circ, 60^\circ)$, $(35^\circ, 55^\circ)$, $(40^\circ, 50^\circ)$, $(45^\circ, 45^\circ)$, $(\theta, 90^\circ - \theta)$ — recording $\sin$ and $\cos$ values to 4 decimal places using either a scientific calculator or the ARES trig table provided. Part B: They calculate the difference ($\sin \theta - \cos(90^\circ - \theta)$) for each row and record the result. Part C: They write a one-sentence observation in their own words about what they notice. Groups are then asked: 'Does this pattern hold for the 38° angle of elevation we used for the water-tank calculation in Lesson 4? Check it.' Learners verify using their calculators.	Distribute the activity sheet (or write the table framework on the board for learners to copy). Say: 'You have 12 minutes to complete Parts A, B, and C with your partner. Use the ARES trig table on page 2 of your sheet, or your calculator — your choice. I am walking around; I will not answer 'Is this right?' — I will only ask you a question back.' [Circulate; when you see a correct difference of 0.0000, crouch and ask: 'What does a difference of zero tell you about those two values?' WAIT 10 seconds for the learner to respond before moving on.] After 12 minutes, pause the class. Cold-call 3 pairs to read their Part C sentences aloud. Write key phrases on the board: 'same value,' 'always equal,' 'swapping $\sin$ and $\cos$.' Then say: 'Now check the 38° case from our tank problem. Does the shortcut hold for our own data?' [WAIT: 2	Structured data table with difference-column verification. The explicit computation of $(\sin \theta - \cos(90^\circ - \theta)) = 0$ transforms an abstract claim into a concrete, checkable arithmetic result. Anchoring the final check to the 38° value from the tank problem ties the pattern directly back to the phenomenon.	Observable indicator: Learners' Part C sentences should use mathematical language connecting $\sin$ and $\cos$ across complementary pairs (e.g., '$\sin 35^\circ$ equals $\cos 55^\circ$ because they are complementary'). A learner who writes 'the numbers are close' rather than 'they are equal' has not yet secured the pattern — prompt: 'What is the exact difference you calculated?' The 38° verification check is a quick whole-class pulse: all pairs should agree that $\sin 38^\circ = \cos 52^\circ$ (≈ 0.6157).

		minutes for pairs to check.] Cold-call 2 pairs for their result.		
Explain Phase  VIDEO: Using complementary angles Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Sine and Cosine of Complementary Angles Practice Source: CK-12  Search ARES for similar readings	Learners now construct the geometrical explanation for the observed pattern. Each learner draws a right-angled triangle in their exercise book with: the right angle at C (the base of the tank), angle θ at A (the student's position on the ground), and angle $(90^\circ - \theta)$ at B (the top of the tank). They label all three sides relative to angle θ first (opposite = BC = height of tank, adjacent = AC = horizontal distance, hypotenuse = AB), then re-label the same sides relative to angle $(90^\circ - \theta)$ at B. Learners write out the ratio definitions for $\sin \theta$ and $\cos(90^\circ - \theta)$ side-by-side and circle the shared fraction. The class then co-constructs a formal two-line proof on the board. To apply the result, learners solve two rapid problems: (i) If $\sin 52^\circ = 0.7880$, write down $\cos 38^\circ$ without a calculator; (ii) A forester in the Mau Forest measures a tree's angle of elevation as 62° from 15 m away — express the height using $\cos 28^\circ$ instead of $\sin 62^\circ$ and confirm the same numerical answer.	Say: 'Put your tables face-down. We know WHAT happens — now let us prove WHY. Draw the water-tank triangle. You have 3 minutes.' [WAIT: 3 minutes. Circulate and check that all three vertices are labelled and the right angle is marked correctly.] Then: 'Label the sides relative to angle θ at A. Write the ratio for $\sin \theta$ as a fraction.' [WAIT: 90 seconds.] 'Now — same triangle, same sides — label the sides relative to angle $(90^\circ - \theta)$ at B. Write the ratio for $\cos(90^\circ - \theta)$ as a fraction.' [WAIT: 90 seconds.] Cold-call one learner to come to the board and write both fractions. Ask the class: 'Are these two fractions the same fraction? Give me a thumbs up or thumbs down — no in-between.' Address any thumbs-down. Say: 'This is why the shortcut works — not magic, just the same sides wearing different names.' Then display the two application problems. Say: 'Solve both independently — silent work, 5 minutes.' [WAIT: 5 minutes.] Cold-call one learner per problem. For the Mau Forest problem, ask: 'Does a forester in the field need to know $\sin 62^\circ$ AND $\cos 28^\circ$, or just one of them? Why does that save effort?' [WAIT: 10 seconds before accepting answers.]	Dual-labelling diagram analysis (a form of Representation Mapping). By labelling the SAME triangle twice — once from each acute angle — learners directly see the structural reason for the identity. The co-constructed board proof converts the pattern from empirical observation to deductive certainty. The Mau Forest application contextualises the algebraic result in a real Kenyan professional scenario.	Observable indicator: In the dual-labelling task, every learner's diagram should show that the side labelled 'opposite to θ' is the same physical segment as the side labelled 'adjacent to $(90^\circ - \theta)$.' A learner who draws two separate triangles has misunderstood — redirect: 'Use only one triangle; just change the angle you are looking from.' In the application problems, the correct numerical answer to the Mau Forest problem is height = $15 \times \cos 28^\circ = 15 \times 0.8829 \approx 13.24$ m (confirming $15 \times \sin 62^\circ \approx 13.24$ m).
Driving Question Board (DQB) Creation  VIDEO: Solving for a side in right triangles with trigonometry	The class gathers around the Driving Question Board. The teacher reads aloud the current un-answered sticky notes from previous lessons — particularly any that asked 'Why do two students at different distances get different angles but the SAME	Hand each learner one green and one yellow sticky note. Say: 'You have 4 minutes — silent writing. Green: something you can now explain about our tank problem that you could NOT explain before today. Yellow: a new question this lesson has opened up.' [WAIT: 4	Driving Question Board with colour-coded sticky notes (Explain / Wonder). This metacognitive routine makes learning progress visible and communal — learners see their own growing understanding mapped on the wall. Moving an old	Observable indicator: A learner's green sticky note should explicitly connect the complementary-angle identity to the water-tank triangle or the two-student scenario (e.g., 'sin of the elevation angle equals cos of the angle at the top of the tank'). Vague notes such as 'I

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Relate ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	height?' and 'Is there a shortcut between sin and cos?' Learners write on green sticky notes at least one thing they can NOW explain that they could not explain before this lesson, and on yellow sticky notes at least one new question this lesson has raised (e.g., 'Does $\sin \theta = \cos(90^\circ - \theta)$ work for angles greater than 90°?' or 'Is there a similar shortcut for tan?'). The class discusses whether the complementary-angle identity helps resolve the two-student consistency puzzle from the anchoring phenomenon.	minutes.] Ask 3 learners to read their green notes and physically place them on the board in the 'Explained' column. Ask 2 learners to read their yellow notes and place them in the 'Still Wondering' column. Then point to the sticky from Lesson 1 or 2 that asked about the two-student consistency: 'Look at this question we posted weeks ago. Can we move it to Explained today? Who wants to make the case?' [WAIT: 10 seconds.] Take 2 volunteers. Confirm or gently correct the explanation, then ceremonially move the sticky to the Explained column.	sticky note to 'Explained' provides narrative closure and motivates continued engagement with the phenomenon.	learned about sin and cos' indicate surface-level processing — the teacher should follow up individually after class. Count the number of yellow 'new question' notes: a rich board at this stage (Lesson 6 of 8) should have both old stickies moving to 'Explained' and new, more sophisticated questions appearing.
Model Building Phase VIDEO: Finding reciprocal trig ratios Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners retrieve their cumulative model drawings from Lesson 5 (a right-angled triangle annotated with the water tank, a student position, θ, opposite, adjacent, hypotenuse, and the formula $h = d \times \tan \theta$). They now add a second layer of annotation in a different colour: (a) label the complementary angle ($90^\circ - \theta$) at the top vertex of the triangle; (b) write the identity $\sin \theta = \cos(90^\circ - \theta)$ as a model note; (c) add an arrow and label showing that 'the height that is OPPOSITE to the student's angle is ADJACENT to the angle at the top' — making the role-swapping visible. Pairs then share their updated models with another pair (Gallery-style: models are placed on desks and pairs rotate) and give one piece of written feedback on a small slip of paper: 'One thing your model clearly shows' and 'One thing to improve.'	Say: 'Take out your cumulative model from last lesson. Use a different colour pen — green, red, or pencil, something different — so the new layer is easy to spot. You have 7 minutes to add the three new annotations I am writing on the board.' [Write the three annotation tasks on the board and leave them there throughout.] [WAIT: 7 minutes. Circulate; prompt any learner who has not added the complementary angle label: 'Where is the other acute angle in the triangle? What is its size if the elevation angle is θ?'] After 7 minutes: 'Stand. Place your model on your desk. Rotate clockwise to the next desk and spend 2 minutes reading that model. Write your feedback slip and leave it on that desk.' [WAIT: 4 minutes for two rotations.] Bring class back together. Ask: 'What is one improvement you saw on someone else's model that you will steal for your own?' [WAIT: 10 seconds.] Cold-call 3 learners.	Cumulative annotated model revision with peer gallery feedback. The multi-colour layering makes the growth of understanding visible in a single artefact across all eight lessons. The gallery rotation embeds peer assessment and forces learners to read and critique mathematical diagrams — a key scientific and engineering practice (Obtaining, Evaluating, and Communicating Information).	Observable indicator: The updated model should contain (a) a correctly sized complementary angle at the top vertex, (b) the written identity $\sin \theta = \cos(90^\circ - \theta)$, and (c) a clear annotation showing the height is simultaneously 'opposite θ' and 'adjacent to $(90^\circ - \theta)$.' A model that only adds the complementary angle label without the side-role annotation has not fully captured the conceptual insight — use the peer feedback slips to surface these gaps. Collect 4–5 models at random at the end of class to review before Lesson 7.

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) During the Observe phase table activity, did learners independently articulate ' $\sin \theta = \cos(90^\circ - \theta)$ ' in their Part C sentences, or did they need teacher scaffolding to name the pattern precisely? If scaffolding was needed, consider providing a sentence stem in future: 'I notice that $\sin$ ____ always equals $\cos$ ____, because ____.' (2) In the Explain phase dual-labelling task, how many learners initially drew two separate triangles instead of re-labelling one? This is a significant conceptual hurdle — plan a brief warm-up at the start of Lesson 7 where learners verbally explain 'same triangle, different viewpoint' to consolidate the idea before the clinometer build. (3) Did the DQB sticky-note activity succeed in moving the two-student consistency question to 'Explained'? If learners' green notes were still vague, the connection between the algebraic identity and the phenomenon is not yet secure — open Lesson 7 with a 3-minute pair discussion: 'If Student A stands 10 m from the tank base and measures 55° elevation, and Student B stands 15 m away and measures 40° elevation, use what you know about $\sin$, $\cos$, and complementary angles to show they must get the same height.' (4) Review the 4–5 collected cumulative models: do all three annotation layers appear in distinct colours? A model with only one colour throughout suggests the learner did not distinguish new from prior knowledge — a metacognitive gap worth addressing individually.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners completed a table of 7 complementary angle pairs (e.g., 35° & 55° , 45° & 45°) and computed $\sin$ and $\cos$ values, finding that the difference $\sin \theta - \cos(90^\circ - \theta) = 0$ for every pair, including the 38° elevation angle used in the water-tank calculation in Lesson 4.
What did I learn?	In a right-angled triangle, the side that is 'opposite' to one acute angle is 'adjacent' to the other acute angle, because the two acute angles are complementary (they sum to 90°). This means $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$ for all acute angles θ — so knowing one ratio immediately gives the other without re-measuring or recalculating.
How does this explain the phenomenon?	The reason two students at different distances from the water tank — measuring different angles of elevation — calculate the same height is that their two elevation angles and the corresponding angle at the top of the triangle are complementary and consistent with the SAME fixed height. The complementary-angle identity shows that switching between $\sin$ and $\cos$ across the two acute angles of the right-angled tank triangle always produces equivalent expressions for the same physical length — confirming the consistency the phenomenon promised.

LESSON 7 (80 minutes): Clinometer Construction & Field Calibration — Building the Measurement Tool

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners design and construct a functional clinometer from locally available materials, calibrate it against known angles, and use it to measure angles of elevation of fixed structures on the school compound — directly preparing them to solve the anchoring phenomenon in Lesson 8.
Knowledge	Learners demonstrate that the tangent ratio links a measured angle of elevation and a known horizontal distance to an unknown vertical height; they explain why two observers at different horizontal distances from the same object must obtain different angles of elevation yet calculate the same height.
Skills	Learners accurately construct a clinometer using a protractor, string, and plumb bob; sight a target at a measured horizontal distance; read and record the angle of elevation; and substitute values into $\tan \theta = \text{opposite/adjacent}$ to calculate the height of a structure without climbing it.
Attitudes	Learners demonstrate a caring and collaborative attitude (Love — KICD value) by sharing materials and taking turns sighting targets, and show self-esteem (PCI) by recognising that a tool they built themselves can replicate the method used by Kenyan surveyors and forest rangers.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	Lessons 1–3 established the three trigonometric ratios and the complementary-angle relationship. Lessons 4–6 applied those ratios to find unknown sides and angles in worked problems. Lesson 7 is the bridge lesson: learners translate abstract ratio knowledge into a physical measuring instrument and verify it against a real structure, consolidating every prior lesson and positioning them to complete the full phenomenon investigation in Lesson 8.
Safety Notes	Learners must not aim the clinometer at the sun. String and scissors should be handled with care. When working outdoors, learners must stay within the designated school compound area and work in pairs so no learner is left alone.

B. LESSON OVERVIEW

Learners construct a simple clinometer (inclinometer) using a printed or hand-drawn protractor, a 20 cm piece of string, a small stone or metal nut as a plumb bob, and a drinking straw as a sighting tube — all materials sourced locally. They first calibrate the device indoors by sighting a wall mark at a known height and measured horizontal distance, checking that $\tan \theta \approx \text{height/distance}$. They then move outdoors in pairs to measure the angle of elevation of a fixed school structure (flag pole, water tank, or classroom wall gable), record the horizontal distance with a tape measure, and calculate the structure's height using the tangent ratio. Pairs at different horizontal distances compare their calculated heights, experiencing the powerful consistency check embedded in the anchoring phenomenon. The lesson closes with a DQB update and a full model revision that integrates the clinometer method into learners' cumulative explanatory model of how surveyors determine heights without climbing.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Finding reciprocal trig ratios Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner studies a diagram showing two students — Amina standing 10 m from the base of the school water tank and Baraka standing 15 m from the same base. Amina's clinometer reads 42° and Baraka's reads 30°. Learners are asked: 'Without any extra information, predict whether Amina and Baraka will calculate the SAME height or DIFFERENT heights. Write your prediction and one sentence of reasoning in your exercise book.' They share predictions with their seat-partner (Think-Pair-Share, 2 minutes), then three volunteers are cold-called to read their reasoning aloud.	Display the diagram and say: 'Look carefully — two people, two different angles, two different distances. I want you to PREDICT before we build anything. Write your answer now — you have 90 seconds, silence please.' [Allow 90 seconds.] 'Now turn to your partner and read what you wrote.' [Allow 2 minutes.] Cold-call three learners by name: 'Wanjiku, read us your prediction.' 'Otieno, do you agree or disagree — why?' 'Fatuma, which prediction do you think is correct?' Do NOT confirm or deny any prediction. Record the three positions on the board under the heading 'Our Predictions'. Say: 'We will return to these predictions after we build and use our clinometers today.'	Think-Pair-Share anchored to the phenomenon: activates prior knowledge of the tangent ratio from Lessons 4–6 and creates cognitive dissonance — the puzzle of different angles yielding the same height motivates the construction activity that follows.	Observable indicator: learner's written prediction references the tangent ratio or the relationship between angle and distance (e.g., 'as distance increases the angle decreases, so tan changes but height stays the same'). Teacher scans 4–5 books during pair-share to check whether Lesson 4–6 ratio knowledge is being recalled.
Observe Phase  VIDEO: Intro to the trigonometric ratios Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Determine and Use the Tangent Ratio (Flexbook) Source: CK-12  Search ARES for similar readings	PART A — Clinometer Construction (25 min, indoors). In groups of three, learners receive a materials pack: one printed protractor sheet (or draw one on card), one 20 cm string, one metal nut (plumb bob), a drinking straw, tape, and scissors. Following a step-by-step instruction card (read aloud by the teacher), they: (1) tape the straw along the straight edge of the protractor; (2) tie the string through the centre-hole of the protractor; (3) attach the nut to the string so it hangs freely as a plumb bob; (4) mark 0° at the plumb-rest position. They test the device by sighting along the straw at a horizontal mark on the wall and confirming the string reads 0°. PART B — Indoor Calibration (10 min). The teacher has pre-marked a red dot on the classroom wall at a height of 1.5 m above the floor. Groups measure exactly 3 m from the wall with a tape measure, sight	During construction: circulate and say — 'Check that your plumb bob hangs freely and the string is not touching the protractor face. If it touches, your readings will be wrong.' After construction, demonstrate the correct sighting technique: 'Close one eye, look through the straw until the target is centred, THEN read the string — not before.' During calibration: after groups record their readings say, 'Everyone freeze — hands up if your clinometer read between 24° and 29°.' [Count and announce.] 'If your reading is outside that range, what part of the construction might have caused the error?' [WAIT 10 seconds.] Cold-call two groups to explain their error. During outdoor work: stand at the base of the target structure and call out, 'Remember — measure horizontal distance along the GROUND, not diagonally. Your tape must be flat.'	Structured inquiry with a built-in consistency check: by requiring two measurements at different distances for the same target, learners generate their own evidence that $\tan \theta = \text{height/distance}$ holds regardless of position — directly mirroring the anchoring phenomenon. The indoor calibration provides an immediate feedback loop against a known answer (26.6°), so learners can diagnose construction errors before going outdoors.	Observable indicators: (1) clinometer plumb bob hangs freely and the device reads within $\pm 3^\circ$ of 26.6° on the calibration target; (2) results table has four entries: two horizontal distances and two corresponding angles for the outdoor target. Teacher collects one results table per group at the end of Phase 2 to scan for systematic errors (e.g., all groups reading 5° low, indicating a straw-alignment issue) before Phase 3 begins.

	the dot, and read the angle of elevation from the clinometer. They calculate the expected angle: $\tan \theta = 1.5/3 = 0.5$, so $\theta = 26.6^\circ$. They record their clinometer reading and compare it to 26.6°. PART C — Outdoor Measurement (20 min). Pairs go outdoors and select one of three pre-assigned targets: (i) the top of the school flag pole, (ii) the highest visible point of the water tank frame, or (iii) the apex of the classroom gable. Each pair measures two different horizontal distances (e.g., 8 m and 12 m), sights the target from each position, and records both angles and both distances in a results table.	Every 5 minutes do a 30-second group-check: 'Show me your results table — I need to see two distances AND two angles recorded.'		
Explain Phase  VIDEO: Finding reciprocal trig ratios Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Back in the classroom, each pair uses their two (distance, angle) data pairs to independently calculate the height of their outdoor target using $\tan \theta = \text{opposite/adjacent}$, rearranged to: $\text{height} = \text{distance} \times \tan \theta$. They record both calculated heights and find the percentage difference between them using $h_1 - h_2 /h_1 \times 100$. Groups that chose the same target compare results across groups. The class discusses: 'Why are our two calculated heights for the same structure almost the same even though we stood at different distances?' Learners write a two-sentence explanation in their exercise books connecting their result to Amina and Baraka's scenario from the Predict phase. Three volunteers read their explanations. The teacher then presents the formal worked example on the board: A student stands 12 m from the base of the Nyahururu water tower frame and measures an angle of elevation of	When learners begin calculating, say: 'You have exactly 8 minutes — both calculations must be complete before we discuss. Work silently.' [Set visible timer.] After 8 minutes: 'Raise your hand if your two calculated heights differ by less than 10%.' [Count.] 'Excellent. Raise your hand if they differ by MORE than 10%.' [Count and note groups.] Say to the high-error groups: 'Don't erase — your data tells us something important about measurement precision. Hold that thought.' During explanation writing, say: 'Your two sentences must include the word TAN and the word DISTANCE — both of them.' [WAIT 15 seconds for writing.] Cold-call three learners: 'Kamau, read your explanation.' 'Does anyone want to add to what Kamau said?' 'Achieng, does your explanation agree with Kamau's — where does it differ?' During the worked example: 'Why is height₂ slightly larger than height₁? Is the building growing?' [WAIT 10	Claim-Evidence-Reasoning (CER) writing: learners make a claim (same height), cite their numerical evidence (two calculated values), and reason using the tangent ratio. This moves them from procedural calculation to conceptual understanding of why the consistency check works — the core explanation of the anchoring phenomenon.	Observable indicators: (1) both calculations correctly apply $\text{height} = \text{distance} \times \tan \theta$ with correct substitution; (2) two-sentence explanation explicitly states that different angles at different distances produce the same height because $\tan \theta$ scales with both simultaneously; (3) percentage difference is calculated (not just the two heights). Teacher circulates and stamps or ticks correct CER sentences as an exit checkpoint before Phase 4.

	35°. A second student stands 20 m away and measures 23°. Calculate the height from both measurements and show they agree. ($\text{height}_1 = 12 \times \tan 35^\circ = 12 \times 0.700 = 8.40 \text{ m}$; $\text{height}_2 = 20 \times \tan 23^\circ = 20 \times 0.4245 = 8.49 \text{ m}$ — small difference due to measurement rounding.) Learners copy and annotate the solution, labelling opposite, adjacent, and hypotenuse on the diagram.	seconds.] 'It is measurement error — this is why surveyors take multiple readings. What did Wanjiku predict at the start of the lesson — was she right?'		
Driving Question Board (DQB) Creation  VIDEO: Finding reciprocal trig ratios Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The class DQB (a poster or section of the board maintained across all 8 lessons) is brought out. Learners review the three columns: WHAT WE WONDERED, WHAT WE NOW KNOW, WHAT WE STILL NEED TO FIND OUT. In pairs, learners write one new sticky note (or board entry) for the KNOW column: a concise statement of what today's clinometer activity proved. They also write one entry for the STILL NEED column: anything remaining before they can fully answer the driving question. Contributions are read aloud and placed on the board. The teacher draws attention to any entry that says 'we still need to measure the actual water tank' and circles it in red, saying: 'That is exactly what Lesson 8 is about — you will take your clinometer outdoors and solve the original problem completely.'	Hand each pair two sticky notes (or ask them to tear a page into two). Say: 'You have 3 minutes — KNOW column: what did today's clinometer activity prove? STILL NEED column: what one thing must happen in Lesson 8 to fully answer our driving question?' [Allow 3 minutes.] Ask four pairs to read their KNOW entries and place them. Ask three pairs to read their STILL NEED entries. After all entries are placed, point to the original driving question written at the top of the DQB: 'How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment without ever measuring it directly?' Ask: 'Has today moved us closer to answering this? Thumbs up / thumbs sideways / thumbs down.' [Scan the room.] 'I see mostly thumbs up — Lesson 8 closes the loop completely.'	DQB revision as cumulative knowledge tracking: making the class's growing understanding visible across all 8 lessons helps learners see their own intellectual progress, consolidates lesson-to-lesson coherence, and creates anticipation for the final lesson — a motivational scaffold for the outdoor investigation.	Observable indicator: KNOW sticky note correctly identifies that $\tan \theta = \text{height/distance}$ and that different (distance, angle) pairs for the same target yield the same height. STILL NEED entry identifies a specific remaining task (e.g., 'measure the actual school water tank height using our clinometer'). Teacher photographs the updated DQB for the lesson record.
Model Building Phase  VIDEO: Finding reciprocal trig ratios Source: Khan Academy (English	Learners open their cumulative explanatory model (a labelled diagram started in Lesson 1 and revised each lesson). Today they add: (a) a drawing of their constructed clinometer with labels — sighting straw, protractor scale, plumb-bob string, angle of	Say: 'Open your model to the most recent version — the one you updated in Lesson 6. You have 10 minutes to add today's three components. Use your results table and your CER sentence as source material.' [Set timer.] After 8 minutes: 'Stop drawing and do	Cumulative model revision (iterative modelling practice from NGSS SEP-2 and SEP-6): each lesson's addition makes the model more complete and more connected to the phenomenon. By Lesson 8, the model will constitute a full explanatory account of the	Observable indicators: (1) clinometer diagram has at minimum three correct labels; (2) right-triangle diagram shows $h = d \times \tan \theta$ with values from today's outdoor measurement substituted; (3) consistency-check arrow is present and annotated; (4) peer-

- US curriculum) Search ARES for similar videos EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum) Search ARES for similar readings	elevation θ; (b) a right-angled triangle diagram showing the water tank scenario with horizontal distance d, height h, and angle θ labelled, and the equation $h = d \times \tan \theta$ written beside it; (c) an arrow from 'Amina at 10 m (42°)' and 'Baraka at 15 m (30°)' both pointing to 'same $h = 9.0$ m' to represent the consistency check; (d) a one-sentence connection: 'This is how Kenyan surveyors mapped the Rift Valley escarpment and how foresters in the Mau Forest measure trees without felling them.' Pairs peer-review each other's models using a three-point checklist on the board: ① clinometer labelled correctly, ② right-triangle equation present, ③ consistency check arrow drawn. Models are signed by the peer reviewer and collected for teacher review.	the peer-review now — use the three-point checklist on the board. Sign your partner's model only if all three points are present. If something is missing, point to it and give your partner 2 minutes to add it.' After peer review: cold-call two learners to hold up their models and describe one addition they made today. Say: 'In Lesson 8, your model will receive its final addition — the actual measured height of our school water tank. Keep your model safe.' Collect models in the exercise books or folders.	driving question — making the learning trajectory visible and meaningful.	review checklist is signed. Teacher uses collected models as a summative data point before Lesson 8 — any learner missing the equation $h = d \times \tan \theta$ receives a targeted prompt at the start of Lesson 8.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the indoor calibration, what proportion of groups obtained a clinometer reading within $\pm 3^\circ$ of the expected 26.6° ? If more than one-third of groups were outside this range, identify whether the error was a construction error (straw misaligned), a reading error (plumb bob not settled before reading), or a measurement error (horizontal distance not measured flat). Plan a 5-minute re-calibration drill at the start of Lesson 8. (2) When learners compared their two calculated heights from different distances, did the percentage differences cluster below 10%? If several groups had large discrepancies, was this because outdoor sighting is harder than indoor sighting, or because the tape measure was held on a slope? Adjust the outdoor measurement protocol for Lesson 8 accordingly. (3) Did the DQB update reveal any persistent misconception — for example, learners believing the angle of elevation is measured from the vertical rather than the horizontal? Address this explicitly in the Lesson 8 opening. (4) Were all learners physically able to participate in the outdoor phase — consider any learners with visual impairments or mobility limitations and plan inclusive roles (e.g., tape-measure holder, data recorder) for Lesson 8. (5) Did today's lesson adequately connect to the Mau Forest / Rift Valley surveying context? If learners seemed disengaged from the real-world connection, bring a printed photograph of a Kenyan forest ranger or land surveyor using a theodolite to open Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners constructed a clinometer from a protractor, string, plumb bob, and drinking straw; calibrated it against an indoor mark at a known height of 1.5 m and horizontal distance of 3 m (expected angle 26.6°); then went outdoors and measured the angle of elevation of a school structure from two different horizontal distances, recording all values in a results table.
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What did I learn?	Learners applied $\text{height} = \text{distance} \times \tan \theta$ to calculate the height of a real school structure from two different (distance, angle) pairs, confirmed that both calculations yielded nearly the same height, and explained in writing (CER format) why the tangent ratio guarantees consistency — directly resolving the core puzzle of the anchoring phenomenon.
How does this explain the phenomenon?	Learners updated their cumulative explanatory model to include a labelled clinometer diagram, a right-triangle equation, and a consistency-check annotation showing that Amina at 10 m (42°) and Baraka at 15 m (30°) both arrive at the same height — and connected this method to how Kenyan surveyors and Mau Forest rangers determine heights without direct measurement.

LESSON 8 (80 minutes): Final Explanation: Solving the Water-Tank Problem and Closing the Explanatory Loop

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate all trigonometric ratio skills (sin, cos, tan) and use them to fully explain the anchoring phenomenon — calculating the height of the school water-tank tower using a measured angle of elevation and a known horizontal distance — without ever measuring the height directly.
Knowledge	Learners know that in a right-angled triangle the sine, cosine and tangent ratios relate any acute angle to the lengths of the opposite, adjacent and hypotenuse sides; they know that sines and cosines of complementary angles are equal; and they know how to select the correct ratio and solve for an unknown side.
Skills	Learners can: (1) identify the correct trigonometric ratio for a given right-angled triangle problem; (2) set up and solve a trigonometric equation to find an unknown side; (3) use a self-made clinometer to measure an angle of elevation outdoors; (4) verify that two observers at different distances obtain the same calculated height — the phenomenon's consistency check; (5) revise their cumulative explanatory model to a complete, final form.
Attitudes	Learners appreciate that mathematics is a precise and powerful tool for solving real-world measurement problems encountered in Kenyan engineering, surveying, forestry and construction; they demonstrate confidence, intellectual honesty and collaborative spirit during the outdoor measurement and peer-checking activities.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	This is the capstone lesson of the sub-strand. Learners carry their clinometers outdoors, measure the angle of elevation of the school water-tank tower from two different distances, then independently calculate its height using tan. They compare their two answers — the consistency check predicted in Lesson 1 — and resolve any discrepancy through peer discussion. Back in class they compare their Lesson-1 initial model with their final model, complete the Driving Question Board, and deliver a written Final Explanation that closes the anchoring phenomenon explanatory loop completely.
Safety Notes	Learners must not look directly at the sun through their clinometer sight-lines. Keep all learners at least 3 m from the base of the water-tank frame. Supervise movement between outdoor measurement stations. Ensure learners return all tools (protractors, strings, weights) to the teacher before the debrief.

B. LESSON OVERVIEW

In this 80-minute final lesson learners move through five structured phases. They begin indoors by revisiting their Lesson-1 predictions and initial models of how the water-tank height could be found (Predict). They then go outdoors in pairs to two marked measurement stations (Observer A at 8 m from the base, Observer B at 12 m from the base) and use their clinometers to record angles of elevation of the top of the school water-tank tower (Observe). Back in class they apply the tangent ratio to independently calculate the height from each station, compare answers, and discuss sources of discrepancy (Explain). They then update their Driving Question Board — marking every question as answered and adding a summary statement — and write a Final Explanation paragraph (DQB Completion). The lesson closes with a side-by-side comparison of the Lesson-1 model and the final cumulative model, a whole-class celebration of the closed explanatory loop, and a teacher-led synthesis connecting the method to Kenyan surveying, the Thika Superhighway overpasses and Mau Forest tree-height measurement (Model Building). Assessment is through the written Final Explanation, the outdoor data sheet and the revised model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners sit in their established pairs. Each learner retrieves their Lesson-1 prediction card (kept in their exercise books) on which they wrote: 'My prediction for how we can find the water-tank height without climbing it is...'. They re-read their prediction silently for 2 minutes, then discuss with their partner: 'How close was my prediction to what we have actually learned over the past seven lessons?' Each pair jots down one thing their Lesson-1 self did not yet know (e.g., 'I did not know which ratio to use') and one thing they now know confidently (e.g., 'I now know $\tan \theta = \text{opposite/adjacent}$ and I can rearrange it'). Two pairs share with the class. The teacher then displays the anchoring phenomenon scenario on the board — 'Observer A stands 8 m from the base and measures an angle of elevation of 42° . Observer B stands 12 m from the base and measures an angle of elevation of 35° . Both must calculate the same tank height. Can they?' — and invites a 60-second silent individual written prediction of the expected height before going outdoors.	Begin by saying: 'Take out your Lesson-1 prediction card. Read it carefully — no writing yet — just read.' Allow 2 minutes of silent reading. Then say: 'Turn to your partner. Tell them: one thing your Lesson-1 self did not know, and one thing you now know confidently. You have 2 minutes.' Circulate and listen; note two strong contrasts to share. After 2 minutes cold-call exactly 2 pairs: 'Aisha and Brian — what did you notice?' and 'Wanjiku and Otieno — what was your pair's contrast?' After each response ask: 'Does anyone want to add to that? Hands down — I'll choose.' Wait 10 seconds and cold-call one more learner. Then display the scenario: 'Here is our phenomenon one final time. Observer A at 8 m, angle 42° . Observer B at 12 m, angle 35° . In your book, predict the height right now — one number, your best guess, 60 seconds.' Circulate and note the range of predictions to revisit after outdoor measurement.	Prediction Revisit and Contrast: learners explicitly compare their naïve Lesson-1 prediction with their current understanding. This metacognitive contrast activates prior knowledge, surfaces residual misconceptions, and creates cognitive readiness for the final measurement task. By committing to a numerical prediction before going outdoors, learners have a personal stake in the outcome.	Observable indicator: every learner has a written numerical prediction in their book before the class moves outdoors. The teacher scans books during circulation — a learner who cannot write any number is flagged for pairing with a stronger peer at the outdoor stations. Listen for pairs who conflate sin and tan or who are unsure which side is 'opposite' — address verbally before moving outside.
Observe Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	The class moves outdoors to the school compound where two stations have been pre-marked with chalk circles on the ground: Station A (8 m from the base of the water-tank tower, measured and marked by the teacher before the lesson) and Station B (12 m from the base). Station distances are written on a ground marker card so	Before leaving the classroom say: 'Safety first: do NOT look at the sun. Stay at least 3 m from the tank base. When I call time, stop measuring immediately.' Outside, assign pairs to stations verbally using the register order — first half to Station A, second half to Station B. Demonstrate clinometer use once at Station A: 'Watch me — I	Authentic Data Collection with Built-In Consistency Check: by collecting real measurements from two positions, learners will experience first-hand that small errors in angle measurement propagate to slightly different calculated heights. This productive messiness — anticipated by the phenomenon design — is itself a	Observable indicator: every learner's data sheet has two rows completed (Station A angle and Station B angle) with units. The teacher checks 5 data sheets during circulation. A learner with an angle greater than 70° or less than 20° at either station has likely misread the clinometer — prompt them to re-measure. Collect data

 EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	learners record them on their data sheets. Working in assigned pairs, Observer A pairs stand at Station A and use their clinometers (protractor + string + small weight, built in an earlier lesson) to sight the top of the tank and read the angle of elevation; they record it. Observer B pairs do the same at Station B. Each learner records independently on a pre-printed data sheet: Station, Horizontal distance (m), Angle of elevation (°). Pairs then swap stations and repeat, so every learner has data from both positions. The teacher circulates, checks clinometer technique (eye level, plumb bob vertical), and prompts: 'Hold the base of the protractor level with your eye. Read the angle where the string crosses the protractor scale.' The outdoor phase takes no more than 12 minutes of measurement; 6 minutes are used to walk out, settle and walk back.	place the flat edge along my line of sight to the top of the tank, I let the string hang freely, and I read the angle here — roughly 42°. Now you try.' Give pairs 4 minutes at their first station. Call: 'Swap stations — move now.' Give 4 minutes at the second station. Call: 'Time. Write your final reading and walk back to class.' As learners measure, circulate and cold-call 3 individuals: 'What angle are you reading?' and 'Is that the angle of elevation or depression — how do you know?' Wait 10 seconds after each question for the learner to answer. Note the spread of readings (typically 40°–44° at Station A and 33°–37° at Station B due to clinometer parallax) — these discrepancies will drive the Explain phase discussion.	sensemaking moment: it shows why surveyors use precise instruments, and why mathematical models are idealisations of real-world measurements. The consistency check (do two different observers get the same height?) was posed as a puzzle in Lesson 1 and is now resolved empirically.	sheets before re-entering the classroom so they cannot be altered retrospectively.
Explain Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Ordering triangle sides and angles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Back in class, learners work individually for 10 minutes to perform two full calculations — one using their Station A data (8 m, their measured angle) and one using their Station B data (12 m, their measured angle) — following these steps: (1) Draw and label the right-angled triangle for the scenario (eye-level height negligible or noted). (2) Identify opposite, adjacent, hypotenuse relative to the angle of elevation. (3) Write the correct ratio equation: $\tan \theta = \text{opposite} / \text{adjacent}$, so $\text{opposite} = \text{adjacent} \times \tan \theta$. (4) Substitute values and calculate height (in metres, to 2 d.p.). (5) Compare the two calculated heights and note the difference. After individual working, pairs	Distribute the data sheets back immediately. Say: 'You have 10 minutes — working alone — to calculate the height of the tank twice, once from each station. Show every step: draw the triangle, label it, write the ratio, substitute, solve. Begin.' Set a visible timer. Circulate and check that learners are using $\tan$ (not $\sin$ or $\cos$) and that they have identified adjacent = horizontal distance. After 5 minutes, pause and say: 'Quick check — hands up if you are using $\tan \theta = \text{opposite over adjacent}$.' If fewer than two-thirds raise hands, write the ratio on the board and say: 'The opposite side is the unknown height. The adjacent side is the horizontal distance. $\tan$ is the ratio	Two-Calculation Consistency Check with Error Analysis: learners perform the same calculation twice with different inputs and expect to arrive at the same answer. The slight discrepancy is not dismissed as failure but treated as a rich sensemaking opportunity — it reveals measurement uncertainty, the importance of precision instruments, and the role of averaging in survey practice. This directly resolves the third puzzling feature of the phenomenon (stated in Lesson 1: 'Two students at different distances should arrive at the same height — a powerful consistency check') and transforms the puzzle into a fully explained result.	Observable indicator: each learner's book contains two fully worked calculations with labelled diagrams, a numerical answer for each station, and a written sentence identifying the source of discrepancy. The teacher marks 6 books during the partner-compare phase. A correct calculation at Lesson-exit level requires: correct ratio selected ($\tan$), correct substitution (adjacent = horizontal distance), correct arithmetic to 2 d.p. Learners who used $\sin$ or $\cos$ are redirected immediately with the prompt: 'Which sides do you know — opposite and adjacent, or opposite and hypotenuse?' Wait 10 seconds for self-correction.

	compare answers (3 minutes), then two pairs share with the class on the board (3 minutes). The teacher leads a 6-minute whole-class discussion on sources of discrepancy (clinometer parallax, not accounting for eye-level height, rounding) and confirms the accepted value using the teacher's own precision measurements. Learners circle their 'best answer' and write a one-sentence explanation of the discrepancy in their books. The teacher connects the calculation explicitly to three Kenyan real-world contexts: surveyors mapping the Kenyan Rift Valley highlands, engineers designing Thika Superhighway overpass clearances, and foresters in the Mau Forest estimating tree heights without felling.	that connects them. Carry on.' After 10 minutes say: 'Compare your two height answers with your partner. Are they the same? If not, why not — write one reason.' Allow 3 minutes. Cold-call 2 pairs to share answers on the board: 'Kamau, write your Station A calculation on the left. Adhiambo, write your Station B calculation on the right.' After both are on the board ask the class: 'Why are these answers slightly different even though they describe the same tank? Think for 10 seconds.' Wait 10 seconds, then cold-call 3 learners. Confirm the accepted height. Then say: 'This is exactly how Kenyan government surveyors work in the Rift Valley — they take readings from multiple positions and average them. The maths is the same maths you just did.'		
Driving Question Board (DQB) Creation  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX) Source: CK-12  Search ARES for similar readings	The class returns to the Driving Question Board — the physical or digital board that has accumulated questions, partial answers and sticky notes across all 8 lessons. Working in groups of four, learners spend 5 minutes reviewing the board: each group is assigned a column of questions and must decide for each question: (a) 'Fully answered — we can explain this completely now', (b) 'Partially answered — we understand it but there is more to learn in later grades', or (c) 'Still open'. Groups place a green sticker (fully answered), yellow sticker (partially answered) or orange sticker (still open) next to each question. Groups then report back in 3 minutes (one sentence per group). In the final 4 minutes, the whole class collaboratively composes a	Say: 'For the last time — look at our Driving Question Board. In groups of four, review your assigned column. For each question, decide: green — fully answered, yellow — partially answered, orange — still open. You have 5 minutes.' Assign columns by pointing to groups. Distribute stickers. After 5 minutes: 'Each group, one sentence — what was the status of your questions?' Cold-call each group. After all groups have reported, say: 'Now, together, we are going to write the official answer to our driving question: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment — without ever measuring it directly? Someone start me a sentence.' Wait 10	Collaborative DQB Closure with Sticker Coding: the sticker-coding activity makes the growth of learner understanding visible and concrete — green stickers show questions that were mysterious in Lesson 1 but are now fully resolved. The collaborative authorship of the DQB Summary Statement requires learners to synthesise, negotiate precise language and commit to a shared explanation, which is the highest level of the sensemaking progression across the unit.	Observable indicator: the DQB Summary Statement is written in every learner's book verbatim. Groups that place orange stickers on questions that were directly addressed (e.g., 'What is the tangent ratio?') are asked to revisit the lesson notes and reconsider — the teacher does not move on until all core questions carry green stickers. The distribution of green/yellow/orange is photographed as a formative record of class-level understanding.

	DQB Summary Statement — 2–3 sentences that directly answer the driving question — which the teacher writes on the board as the class dictates it. Each learner copies this statement into their book as the official answer to the driving question.	seconds, then cold-call one learner. Build the statement collaboratively, pressing for precision: 'Can we make that more specific? What ratio did we use? What did the horizontal distance represent?' Write the final 2–3 sentence statement in large text on the board. Say: 'Copy this into your book. It is your Final Answer.'		
Model Building Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Number opposites challenge Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner opens their book to their Lesson-1 initial explanatory model (the diagram or description they drew in Lesson 1 to explain how the water-tank height could be found). They place it beside a blank page. Working individually for 8 minutes, they draw their Final Explanatory Model — a fully labelled diagram of the water-tank scenario showing: the right-angled triangle formed by the tower height, horizontal ground distance and line of sight; the angle of elevation θ at the observer's position; the labels 'opposite = h (height)', 'adjacent = d (horizontal distance)', 'hypotenuse = line of sight'; the equation $\tan \theta = h/d$; the rearranged form $h = d \times \tan \theta$; and a worked numerical example using Station A values. Below the diagram, learners write a 3–5 sentence Final Explanation paragraph that answers the anchoring phenomenon completely. They then draw a double-headed arrow between the Lesson-1 model and the Final Model and annotate it: 'What changed: ...' with at least three specific additions (e.g., 'I now know the name of the ratio', 'I can now set up the equation', 'I understand why two observers get the same answer'). The lesson closes with the teacher inviting 2	Say: 'Open to your Lesson-1 model. Look at it carefully for 30 seconds. Now turn to a fresh page and spend 8 minutes drawing your Final Explanatory Model. It must include: the labelled triangle, the three sides named, the $\tan$ equation, the rearranged form, and a worked example. Go.' Circulate and check that labels use correct terminology (opposite, adjacent, hypotenuse) and that the equation is written in both forms. After 8 minutes say: 'Now draw a double-headed arrow between your Lesson-1 model and your Final Model. Write at least three things that changed. You have 3 minutes.' After 3 minutes, say: 'Who would like to read their Final Explanation paragraph?' Wait 5 seconds, choose 1 volunteer, then cold-call 1 more: 'Thank you. Now I am choosing — Mwangi, please read yours.' After both readings, ask the class: 'What did both explanations have in common? Think for 10 seconds.' Wait 10 seconds, cold-call 2 learners. Close with: 'What you have done today is exactly what Kenyan surveyors do when they map the Rift Valley highlands. What foresters in the Mau Forest do when they estimate tree heights for conservation. What KeNHA engineers do when they design	Lesson-1 vs Final Model Comparison (Cumulative Model Revision): explicitly placing the naïve initial model beside the final model makes conceptual growth visible and personally meaningful. Annotating 'what changed' requires learners to articulate the specific knowledge gains of the unit — a powerful metacognitive consolidation move. The Final Explanation paragraph is the unit's capstone assessment artefact and directly answers the anchoring phenomenon, closing the explanatory loop that was opened in Lesson 1.	Observable indicator: the Final Explanatory Model contains all six required elements (labelled triangle, three sides named, $\tan$ equation in two forms, worked numerical example, Final Explanation paragraph, annotated comparison arrow). The teacher collects these books at the end of the lesson for summative marking against the KICD assessment rubric descriptors: 'Meets Expectations' requires all six elements present and the calculation correct; 'Exceeds Expectations' additionally requires the paragraph to reference real-world application and to explain the consistency check between two observers. Any learner whose paragraph does not reference the horizontal distance as the adjacent side is given written feedback before the next lesson.

	volunteers to share their Final Explanation paragraphs aloud (3 minutes), followed by a 4-minute teacher synthesis that explicitly names the NGSS Science and Engineering Practice of 'Using Mathematics and Computational Thinking' and connects the sub-strand to surveying careers in Kenya, the Kenya National Highways Authority, and Kenyan forestry services.	overpass clearances on the Thika Superhighway. You have unlocked a tool that professionals use every day — and you built it from a water tank in your own school compound.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did every learner successfully select tan (not sin or cos) as the correct ratio for the water-tank scenario — if not, what misconception persisted and how will I address it in the next sub-strand? (2) Were the two calculated heights (from Stations A and B) close enough to confirm the consistency check, or did clinometer errors produce results so different that learners lost confidence in the method? If the latter, how might I improve the outdoor measurement protocol next year — perhaps pre-marking stations more precisely or demonstrating clinometer calibration more carefully? (3) Did the Driving Question Board closure feel genuine — did learners own the summary statement, or were they copying passively? (4) Were the Lesson-1 vs Final Model comparisons detailed enough to demonstrate real conceptual growth? (5) Which learners were still unable to complete a full tan calculation independently by the end of the Explain phase — these learners need targeted support in subsequent geometry sub-strands. (6) Was the connection to Kenyan professional contexts (KeNHA, Mau Forest foresters, Rift Valley surveyors) meaningful to learners, and did it shift their attitude towards the utility of trigonometry?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners used their self-built clinometers at two marked outdoor stations (8 m and 12 m from the school water-tank tower base) to measure and record the angle of elevation of the top of the tank from each position.
What did I learn?	Learners learned that the tangent ratio ($\tan \theta = \text{opposite/adjacent} = \text{height/horizontal distance}$) can be rearranged to $h = d \times \tan \theta$, allowing the unknown height of the tank to be calculated precisely from a known horizontal distance and a measured angle of elevation — without ever climbing the tank or measuring its height directly. They also learned that two observers at different distances should calculate the same height, providing a built-in consistency check, and that small discrepancies arise from measurement uncertainty rather than errors in the mathematics.
How does this explain the phenomenon?	Learners explained the complete anchoring phenomenon: the height of the school water-tank tower can be found by standing at a measured horizontal distance from its base, measuring the angle of elevation of its top with a clinometer, and applying $h = d \times \tan \theta$. This is the same method used by Kenyan surveyors in the Rift Valley highlands, KeNHA engineers designing Thika Superhighway overpasses, and foresters in the Mau Forest estimating tree heights for conservation — confirming that trigonometry is a powerful, universally applicable tool for finding unknown heights and distances without direct measurement.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.